

Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued. . .]

- Appendix on s -wave SC ghost simulations on HM
- Set ComputerModern font in all CairoMakie figures

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Introduction to the Extended Hubbard Model

A major breakthrough in Condensed Matter Physics was the discovery of superconductivity in two-dimensional copper-oxide based compounds (often referred to as “cuprates”). Until 1986, when Bednorz and Muller found evidence of a superconducting transition in Ba-La-Cu-O systems [5], superconductivity had been observed up to critical temperatures (T_c) around 30 K. The new class of copper oxides rapidly took over as highest- T_c class of superconductors, dramatically increasing critical temperatures to around 130 K at ambient pressure. Today, the highest observed value is $T_c \approx 151$ K at ambient pressure on the cuprate $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_{8+\delta}$ [6]. It is worth mentioning that this result was obtained in recent months after almost 30 years from the previous record.

Cuprates Physics is particularly interesting and surprising also considering their very poor conducting properties at normal phase. Copper oxides are basically ceramics, highly prone to establishing insulating antiferromagnetic phases. Antiferromagnetism and superconductivity are apparently antithetical orderings: the first arises from electronic repulsion, while the second from attractive interactions [7]. It is widely observed, however, how in cuprates the two phases can compete **and even coexist** [8].

Moreover, cuprates superconduct through a new, not-yet understood mechanism. Classical Bardeen-Cooper-Schrieffer (BCS) theory of superconductivity describes the arising of a phonon-induced effective electron-electron attractive interaction as a source of so-called “conventional superconductivity”. It is the interplay of electrons with lattice vibrations that couples them in Cooper pairs. The same theory fails completely in explaining this new “unconventional superconductivity”. All of these observation have gathered, in years, a lot of interest in these materials.

Such complex materials are surprisingly well described in many features by the simple (but rich) single-band Hubbard model (HM) [8],

$$\hat{H}_{\text{HM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad t, U > 0 \quad (1.1)$$

being t the hopping energy and U the on-site repulsion. Sum is intended on the spin index $\sigma \in \{\uparrow, \downarrow\}$ and the sites i of a two-dimensional square lattice. The notation $\langle ij \rangle$ indicates nearest neighbours (NNs). Recent developments [9] have shown, however, that a more accurate description of these materials could be obtained by adding a non-local electron-electron attraction. This leads us to a widely studied modified version of Eq. (1.1),

$$\hat{H}_{\text{EHM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \quad t, U, V > 0 \quad (1.2)$$

commonly referred to as “Extended Hubbard model” (EHM). The difference with the conventional HM is the presence of a NNs attractive interaction, independent of site and spin indices. In this thesis, we study the insurgence of antiferromagnetism and superconductivity in the EHM due to the presence of this new interaction. To do so, we use numerical Hartree-Fock (HF) methods.

This introductory chapter is organised as follows: **[To be continued...]**

Chapter 2

Phase transitions and SSB in EHM

The general idea of this project is to apply HF methods to the EHM, derive analytically the approximated quadratic hamiltonian of Eq. (??) and then apply the iHF procedure sketched in Sec. ?? to extract the self-consistent values of the various HFPs. One of the advantages of using iHF methods is that we can impose rather simply the symmetries of the HF approximated wavefunction.

This is most useful when dealing with competing phases. A phase transition is basically described by some spontaneous symmetry breaking (SSB) and the rising to a non-zero value of an order parameter. We will deal with three phases:

1. the normal phase, where no symmetry is broken;
2. the antiferromagnetic phase, where we allow for spin density to be non-uniform across sites;
3. the superconducting phase, where we do not conserve particle number.

Each phase is discussed in a separate chapter. Many other “phases” are possible, based on which symmetries are respected and which are not.

For each phase, we will extract the best approximation to the true ground state wavefunction. In computational terms, we will find three solutions to the general self-consistency problem of Eq. (??). In the end we want to compare the free energy of these three solutions, assessing which one is the most stable. Any HF approximation, being by construction not the true ground state, overestimates energy. Thus, comparing energies of our solutions we get a hint of the features of the true ground state.

2.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model introduced in Eq. (1.2)

$$\hat{H}_{\text{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\hat{H}_t} + \underbrace{U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\hat{H}_U} - \underbrace{V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'}}_{\hat{H}_V}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\text{EHM}}$. We start by the lattice symmetries.

2.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. We indicate with L_x the number of sites in direction x , and L_y in direction y . The hamiltonian of Eq. (1.2) respects three fundamental lattice symmetries:

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as $T_1^{(x)}$, $T_1^{(y)}$. It holds:

$$T_n^{(\ell)} \supset T_{kn}^{(\ell)} \quad k \in \mathbb{N} \quad (2.1)$$

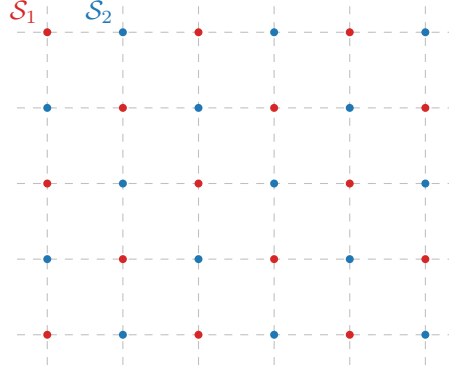


Figure 2.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

A shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n -sites shifts, thus, implies invariance also for kn -sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group C_4 , containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

$$C_n \supset C_{n/k} \quad k, n/k \in \mathbb{N} \quad (2.2)$$

Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$I_2 \equiv C_2 \quad (2.3)$$

In 2D space, to rotate by π around the origin and to mirror the position of a point are the same operation.

The planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v, w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1, \mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \implies (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 2.1. Two sites are “connected” if linked by a non-zero hopping operator. Which is, if they are NNs. Note that one-site translations leave the hamiltonian invariant but exchange the two sublattices, while two-sites translations don’t.

2.1.2 Spin and charge symmetries

The conventional HM shows a global $U(2)$ symmetry, extended to $SU(2) \times SU(2)/\mathbb{Z}_2 = SO(4)$ symmetry at half-filling on a bipartite lattice [14], such as the square lattice. For the EHM, we limit to prove that we can identify a $U(2) \subset SO(4)$ global symmetry.

Gauge symmetry. The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma} \rightarrow e^{-i\eta} \hat{c}_{i\sigma} \quad \hat{c}_{i\sigma}^\dagger \rightarrow e^{i\eta} \hat{c}_{i\sigma}^\dagger \quad \eta \in \mathbb{R} \quad (2.4)$$

To prove this, it suffices to note that

$$\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \rightarrow e^{i\eta} \hat{c}_{i\sigma}^\dagger e^{-i\eta} \hat{c}_{j\sigma'} = \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'} \quad (2.5)$$

As a consequence, any product of fermionic operators with an equal number of annihilations and creations is gauge-symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U, \hat{H}_V$ are invariant. As a consequence, the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma}$$

is a good quantum number. The symmetry is described by the transformation of Eq. (2.4), which is a pure phase shift by η . The relative group is $U^c(1)$, with a superscript “c” indicating “charge”.

Global spin symmetry The EHM is described by three operators: NN hopping, independent of σ ; local repulsion between opposite-spin densities; non-local attraction, independent of σ . If we rotate all spins by an equal amount, the hamiltonian must remain identical. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H}_{\text{EHM}} - \mu\hat{N}$ must exhibit a global $SU(2)$ symmetry, related to the rotation in 3D space of all spins altogether.

Let us prove this rigorously. Let the on-site spinor:

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix} \quad (2.6)$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \quad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \quad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

We define the local spin operator as:

$$\hat{S}_i^\alpha \equiv \frac{1}{2} \hat{\Psi}_i^\dagger \tau^\alpha \hat{\Psi}_i \quad [\hat{S}_j^\alpha, \hat{S}_j^\beta] = i \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{S}_j^\gamma \quad (2.7)$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. The $\mathfrak{su}(2)$ spin algebra is actually respected:

Proof 2.1 (Spin operators respect the $\mathfrak{su}(2)$ algebra).

To prove that $\mathfrak{su}(2)$ algebra is respected, we must compute all possible commutators. This involves many very similar calculations. For the sake of conciseness, we sketch the proof for $\alpha = x, \beta = y$:

$$\begin{aligned} [\hat{S}_j^x, \hat{S}_j^y] &= \frac{1}{4} \left[\hat{\Psi}_j^\dagger \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}_j^\dagger \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j \right] \\ &= \frac{1}{4} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}, -i \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} + i \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] \\ &= \frac{i}{4} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}, \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] - \frac{i}{4} \left[\hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}, \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \right] \\ &= \frac{i}{2} \left[\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}, \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \right] \\ &= \frac{i}{2} \left(\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow} \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= \frac{i}{2} \left(-\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} + \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} \hat{c}_{j\downarrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= \frac{i}{2} \left(\hat{c}_{j\uparrow}^\dagger \hat{c}_{j\uparrow} - \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} \right) \\ &= i \hat{S}_j^z \end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}_{j\sigma'}^\dagger\} = \delta_{ij}\delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other commutators, giving in the end the $\mathfrak{su}(2)$ spin algebra.

Let the raising and lowering operators:

$$\hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2} \quad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i \in \mathcal{L}} \hat{S}_i^\alpha \quad [\hat{S}^\alpha, \hat{S}^\beta] = i \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2} \sum_{i \in \mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \quad \hat{S}^+ \equiv \frac{1}{2} \sum_{i \in \mathcal{L}} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \quad \hat{S}^- \equiv (\hat{S}^+)^\dagger$$

The EHM hamiltonian and the number operator commute with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [14, 27],

$$[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}] = 0 \quad [\hat{N}, \hat{\mathbf{S}}] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold,

$$[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}] = 0 \quad (2.8)$$

Let us give a formal proof.

Proof 2.2 (The non-local attraction $\hat{H}_V$ is symmetric under global spin rotations.).

The proof is organised through a sequence of results:

1. The non-local attraction can be written as:

$$\begin{aligned} \hat{H}_V &= V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \\ &= V \sum_{\langle ij \rangle} \hat{\Psi}_i^\dagger \hat{\Psi}_i \hat{\Psi}_j^\dagger \hat{\Psi}_j \end{aligned} \quad (2.9)$$

having we used $\hat{\Psi}_i^\dagger \hat{\Psi}_i = \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} + \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow}$.

2. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp(-i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}) \quad \text{where} \quad \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

Spinors transform as:

$$\hat{R}(\boldsymbol{\theta}) \hat{\Psi}_i \hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta}) \hat{\Psi}_i$$

with $\mathcal{R}(\boldsymbol{\theta}) \in \text{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau}\right) \quad \text{where} \quad \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \quad (2.10)$$

Let us prove this. Let $\hat{A} = i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$

The first order of the expansion gives:

$$\begin{aligned}
[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i] &= i \sum_{\alpha} \theta_{\alpha} [\hat{S}^{\alpha}, \hat{\Psi}_i] \\
&= -\frac{i}{2} \sum_{\alpha} \theta_{\alpha} \tau^{\alpha} \hat{\Psi}_i \\
&= -\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \hat{\Psi}_i
\end{aligned}$$

We omitted some purely algebraic passages. The second order:

$$\begin{aligned}
[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i]] &= [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, -\frac{i}{2}(\boldsymbol{\theta} \cdot \boldsymbol{\tau}) \hat{\Psi}_i] \\
&= -\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} [i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i] \\
&= \frac{1}{4} (\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2 \hat{\Psi}_i
\end{aligned}$$

and so on. Collecting all terms,

$$\begin{aligned}
\hat{R}(\boldsymbol{\theta}) \hat{\Psi}_i \hat{R}(\boldsymbol{\theta})^{\dagger} &= \hat{\Psi}_i - \frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \hat{\Psi}_i + \frac{1}{2! \times 4} (\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2 \hat{\Psi}_i + \dots \\
&= \sum_{k=1}^{\infty} \frac{1}{k!} \left(-\frac{i}{2} \boldsymbol{\theta} \cdot \boldsymbol{\tau} \right)^k \hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta}) \hat{\Psi}_i
\end{aligned}$$

In last passage, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (2.10).

3. Then, the group acts on the product $\hat{\Psi}_i^{\dagger} \hat{\Psi}_i$ as

$$\begin{aligned}
\hat{R} \hat{\Psi}_i^{\dagger} \hat{\Psi}_i \hat{R}^{\dagger} &= \hat{R} \hat{\Psi}_i^{\dagger} \hat{R}^{\dagger} \hat{R} \hat{\Psi}_i \hat{R}^{\dagger} \\
&= \hat{\Psi}_i^{\dagger} \mathcal{R}^{\dagger} \mathcal{R} \hat{\Psi}_i \\
&= \hat{\Psi}_i^{\dagger} \hat{\Psi}_i
\end{aligned}$$

being $\mathcal{R} \in \text{SU}(2)$. Recalling Eqns. (2.9), we conclude that $\hat{H}_V$ is $\text{SU}(2)$ symmetric – which means, it does not change under global spin rotations. This fact implies Eq. (2.8).

The EHM is symmetric under global spin rotations. Note that $\hat{H}_V$ is actually symmetric under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. The same holds for $\hat{H}_U$. It is the kinetic operator that makes the EHM globally (and not locally) $\text{SU}(2)$ symmetric.

The $U^c(1)$ gauge symmetry and the $\text{SU}(2)$ spin rotations symmetry can be combined,

$$U^c(1) \otimes \text{SU}(2) = \text{U}(2)$$

We will refer to $\text{U}(2)$ symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis z ,

$$U^z(1) \subset \text{SU}(2)$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the “real” z axis, perpendicular with the lattice plane. Any given direction works. Inverting our logic, we can build the $\text{SU}(2)$ generators, starting from $\hat{S}^z$, after we have chosen the direction z . In terms of commutation relations, a given operator can respect $U^z(1)$ but not $\text{SU}(2)$: in that case, it will commute to zero with $\hat{S}^z$, and not the other spin operators. A graphic rendition of this SSB is given in Fig. 2.2, representing for a given vector $\mathbf{m}$ the passage

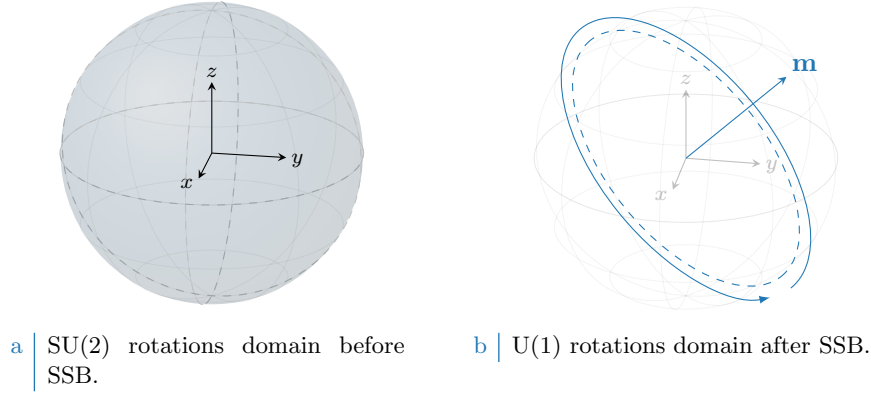


Figure 2.2 | Graphic representation of the concept of $SU(2) \rightarrow U^z(1)$ SSB. “Before” SSB, we have full symmetry under any possible rotation (solid color sphere, left panel). “After” SSB we have symmetry only for rotations around some vector $\mathbf{m}$ (dashed tilted circumference, right panel).

from full $SU(2)$ rotational symmetry around any direction to reduced $U(1)$ rotational symmetry around just one direction.

2.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. Let us prove this. Consider the generic unitary particle-hole (PH) transformation

$$P : \hat{c}_{i\sigma} \rightarrow e^{i\eta_{i\sigma}} \hat{h}_{i\sigma}^\dagger \quad (2.11)$$

being in general η_i a phase dependent on the fermionic quantum state ($i\sigma$). Inserting the above transformation in the EHM, we get

$$P\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}_{j\sigma}^\dagger + U \sum_i \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\uparrow}^\dagger - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}^\dagger$$

For the two interactions $\hat{H}_U, \hat{H}_V$ the phase factors cancelled out. For a specific choice of the η_i phases,

$$P : \begin{cases} \hat{c}_{i\sigma} \rightarrow \hat{h}_{i\sigma}^\dagger & \text{for } i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \rightarrow -\hat{h}_{i\sigma}^\dagger & \text{for } i \in \mathcal{S}_2 \end{cases} \quad (2.12)$$

the above hamiltonian is mapped onto the EHM of (1.2),

$$P\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma} + U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow}^\dagger \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}$$

The equality $P\{\hat{H}\} = \hat{H}$ expresses PHS. This holds as long as we work at half-filling.

Proof 2.3 (PHS at half filling, with η_i given by Eq. (2.12)).

This proof is an extension of the argument carried out for the conventional HM [14]. For the hole operators, Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned} \{\hat{h}_{i\sigma}, \hat{h}_{j\sigma'}^\dagger\} &= e^{i(\eta_{i\sigma} - \eta_{j\sigma'})} \{\hat{c}_{i\sigma}^\dagger, \hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_{i\sigma} - \eta_{j\sigma'})} \delta_{ij} \delta_{\sigma\sigma'} \\ &= \delta_{ij} \delta_{\sigma\sigma'} \end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma}\hat{h}_{i\sigma}^\dagger = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ . It follows, for quartic interactions

$$\begin{aligned} \hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger\hat{h}_{i\sigma}^\dagger &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{i\sigma}^\dagger\hat{h}_{j\sigma'}^\dagger \\ &= \hat{h}_{i\sigma}\hat{h}_{i\sigma}^\dagger\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}\hat{h}_{j\sigma'}^\dagger\delta_{ij}\delta_{\sigma\sigma'} \\ &= -\hat{h}_{i\sigma}^\dagger\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger + \hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}\hat{h}_{j\sigma'}^\dagger\delta_{ij}\delta_{\sigma\sigma'} \\ &= \hat{h}_{i\sigma}^\dagger\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}\hat{h}_{j\sigma'}^\dagger\delta_{ij}\delta_{\sigma\sigma'} \\ &= -\hat{h}_{i\sigma}^\dagger\hat{h}_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger - \left(\hat{h}_{i\sigma}\hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\ &= \hat{h}_{i\sigma}^\dagger\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}^\dagger\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}_{j\sigma'}^\dagger - \hat{h}_{i\sigma}^\dagger\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \quad (2.13) \end{aligned}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$\hat{H}_t = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}$$

Apply the PH transform of Eq. (2.11)

$$P\{\hat{H}_t\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}_{j\sigma}^\dagger$$

Anti-commutate the two fermionic operators, and exchange site labels $i \leftrightarrow j$. We get:

$$P\{\hat{H}_t\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma} + \eta_{j\sigma} + \pi)} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma}$$

To get $P\{\hat{H}_t\} = \hat{H}_t$, a necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} \stackrel{!}{=} (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z} \quad (2.14)$$

valid for NN sites. This requirement on the phases η_i gives Eq. (2.12).

Local repulsion. Substitute Eq. (2.13) in $\hat{H}_U$,

$$\begin{aligned} P\{\hat{H}_U\} &= U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\uparrow}^\dagger \\ &= U \sum_{i \in \mathcal{L}} \hat{h}_{i\uparrow}^\dagger \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} + U \sum_{i \in \mathcal{L}} (\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U \sum_{i \in \mathcal{L}} (1) \\ &= U \sum_i \hat{h}_{i\uparrow}^\dagger \hat{h}_{i\downarrow}^\dagger \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} + U(\hat{N} - L_x L_y) \end{aligned}$$

At half filling, $\langle \hat{N} \rangle = L_x L_y$. The unitary map of Eq. (2.11), independently of the condition of Eq. (2.14), leaves the hamiltonian identical at half-filling.

Symmetry	Operations
$T_1^{(x)} \otimes T_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$U^c(1)$	Charge-conserving global phase shifts
$SU(2)$	Three-dimensional rotations in global spin space

Table 2.1 | Model symmetries and relative group operations. Note that $U^c(1)$ represents the gauge symmetry linked to charge conservation.

Non-local attraction. Proceed identically for $\hat{H}_V$,

$$\begin{aligned}
P\{\hat{H}_V\} &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'}^\dagger \hat{h}_{i\sigma}^\dagger \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'}^\dagger \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (1) \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma}^\dagger \hat{h}_{j\sigma'}^\dagger \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - 2zV(\hat{N} - L_x L_y)
\end{aligned}$$

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}
\sum_{\langle ij \rangle} (1) &= \sum_{i \in \mathcal{L}/2} z \\
&= \frac{L_x L_y}{2} \times z
\end{aligned} \tag{2.15}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. We conclude that the entire EHM exhibits PHS at half filling, if P is defined as in Eq. (2.12).

As for the conventional HM [14], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 2.1 all identified symmetries.

2.2 Symmetry breaking channels in real space

In many-body Physics, a phase transition is described by some order parameter rising from zero to a non-zero value. This is associated with some spontaneous symmetry breaking (SSB). In order to simulate different phases, we need a way to distinguish them based on which symmetries they break (or not). We also need a way to work computationally in a given symmetry sector. HF methods, by reducing the hamiltonian to a non-interacting approximation, offer a simple framework to these ends. Indeed, once we identified all symmetries of a given phase, we just impose them on our approximation of the full hamiltonian. The found ground-state approximation will entail these symmetries by construction.

Recall now the discussion of the HF method of Sec. ???. The ending point were Eqns. (??), (??), which identify the self-consistent single-particle fields which approximate the original quartic hamiltonian. In order to get there, we applied Wick's theorem to the quartic part of the hamiltonian and derived the most general expressions of $\mathbf{h}$. In next sections here apply Wick's theorem to the quartic operators of $\hat{H}_{\text{EHM}}$ and discuss the symmetry properties of the resulting decomposition.

Before starting, let us define precisely the three phases we will investigate in terms of symmetries:

- Normal phase: no broken symmetry.
- Antiferromagnetic phase: simultaneous $SU(2) \xrightarrow{\text{SSB}} U^z(1)$ and $T_1^{(x)} \otimes T_1^{(y)} \xrightarrow{\text{SSB}} T_2^{(x)} \otimes T_2^{(y)}$. Among all possible realisations of antiferromagnetism, we realise a SDW with alternating distribution of spin densities;

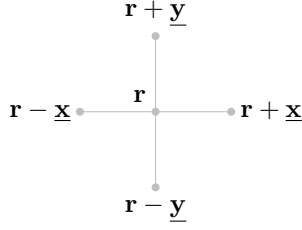


Figure 2.3 | Schematic representation of the four NNs of a given site i for a planar square lattice.

- Superconducting phase: simultaneous $U^c(1) \xrightarrow{\text{SSB}} 0$ and $C_4 \xrightarrow{\text{SSB}} C_2$. We investigate anisotropic superconductivity, allowing for the particle number to be not conserved and for nematic effects.

Nematic effects are observed in unconventional superconductors [23], and this is the reason of our interest in the possibility of rotational SSBs. We will theoretically discuss the nematic SSB $C_4 \rightarrow C_2$ also for normal and AF phases, as if discrete rotational symmetry was broken. As we will see, while it not trivial to exclude it *a priori*, it will actually turn out that the HF ground state preserves C_4 in both cases.

We now move to a more detailed discussion of each hamiltonian contribution. In next sections, we switch from the site index i to the vector notation $\mathbf{r}$, indicating the site 2D position on the lattice

$$i \rightarrow \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1, 0) \quad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 2.3.

2.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is the quartic operator of the conventional HM,

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \end{aligned}$$

In order to obtain the self-consistent values of $h_{\alpha\beta}^{+-}$, $h_{\alpha\beta}^{++}$ we need to compute an expectation value of the full hamiltonian. Wick's theorem guarantees that, for the wavefunctions we are using, the expectation value of a quartic fermionic operator is the sum of all fully contracted terms. Recalling the nomenclature of Eq. (??),

$$\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle = \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}} \quad (2.16)$$

where the equal sign holds as long as we perform the averaging operation on the subspace of gaussian states. Now we discuss the above EVs separately, identifying if any can be neglected based on selection rules in a given phase.

Hartree term. The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle$$

Recalling the groups of Tab. 2.1, we see that the density operator preserves all symmetries. Single-site density operators commute with spin operators and are gauge-invariant. Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no selection rule applies, and the above EVs are in principle non-zero.

Fock terms. For the Fock part, notice that

$$\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} = \hat{S}_{\mathbf{r}}^+$$

being $S_{\mathbf{r}}^+$ the local raising operator. From $\mathfrak{su}(2)$ algebra we have:

$$[\hat{S}_{\mathbf{r}}^+, \hat{S}^\alpha] \neq 0 \quad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, these EVs must be zero. Gauge symmetry gives no selection rules, due to Eq. (2.5).

Cooper terms. The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (2.4) to $\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$

$$\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \rightarrow e^{2i\eta} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$$

to conclude that this operator is in general not gauge invariant. It is instead $SU(2)$ -symmetric. Let us provide a proof:

Proof 2.4 (Local Cooper terms are $SU(2)$ -symmetric).

We must compute the commutators of $\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$ with $\hat{S}^z, \hat{S}^+$:

$$\begin{aligned} [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^z] &= \left[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} (\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow}) \right] \\ &= \frac{1}{2} [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] \\ &= \frac{1}{2} [\hat{c}_{\mathbf{r}\uparrow}^\dagger, \hat{n}_{\mathbf{r}\uparrow}] \hat{c}_{\mathbf{r}\downarrow}^\dagger - \frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger [\hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{n}_{\mathbf{r}\downarrow}] \end{aligned}$$

Since:

$$\begin{aligned} [\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] &= \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} - \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma}^\dagger \\ &= -\hat{c}_{\mathbf{r}\sigma}^\dagger (1 - \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma}) \\ &= -\hat{c}_{\mathbf{r}\sigma}^\dagger \end{aligned} \tag{2.17}$$

we have:

$$[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^z] = -\frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger + \frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger = 0$$

Hence, $\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger$ is $U^z(1)$ -symmetric.

Moreover, considering the raising operator:

$$\begin{aligned} [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{S}^+] &= \left[\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} \hat{c}_{\mathbf{r}'\uparrow}^\dagger \hat{c}_{\mathbf{r}'\downarrow} \right] \\ &= [\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}] \\ &= [\hat{c}_{\mathbf{r}\uparrow}^\dagger, \hat{c}_{\mathbf{r}\uparrow}^\dagger] \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} - (\hat{c}_{\mathbf{r}\uparrow}^\dagger)^2 [\hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{c}_{\mathbf{r}\downarrow}] \\ &= 0 \end{aligned}$$

where we used the facts that every operator commutes with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a $SU(2)$ -preserving phase.

In Tab. 2.2 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (2.16), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 2.2 | Summary of selection rules for the Wick's decomposition of Eq. (2.16) of the local repulsion $\hat{H}_U$. An empty entry indicates no selection rule.

2.2.2 Non-local attraction

In order to analyse similarly the non-local attraction $\hat{H}_V$, it is useful to decompose the operator in spin channels:

$$\begin{aligned}\hat{H}_V &\equiv \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}} \\ &= \hat{H}^{(\text{s.s.})} + \hat{H}^{(\text{o.s.})}\end{aligned}\quad (2.18)$$

being

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'}$$

Let us start from the opposite spin sector. On the square lattice (as for any bipartite lattice) NNs belong to different sublattices. This implies that any NNs sum can be rewritten as:

$$\sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} f(\mathbf{r}, \mathbf{r}') = \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\boldsymbol{\delta} \in \{\mathbf{x}, \mathbf{y}\}} f(\mathbf{r}, \mathbf{r} + \boldsymbol{\delta}) \quad (2.19)$$

with f some function. We are using the notation of Fig. 2.3. The symbol “ $\mathcal{L}/2$ ” indicates one of the two sublattices $\mathcal{S}_1, \mathcal{S}_2$ represented in Fig. 2.1. We can rewrite:

$$\hat{H}_V^{\sigma\sigma'} = -V \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\boldsymbol{\delta} \in \{\mathbf{x}, \mathbf{y}\}} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'})$$

Now, if we choose $\mathcal{L}/2 = \mathcal{S}_1$, this gives:

$$\begin{aligned}\hat{H}_V^{\uparrow\downarrow} &= -V \sum_{\mathbf{r} \in \mathcal{S}_1} \sum_{\boldsymbol{\delta} \in \{\mathbf{x}, \mathbf{y}\}} (\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}) \\ \hat{H}_V^{\downarrow\uparrow} &= -V \sum_{\mathbf{r} \in \mathcal{S}_1} \sum_{\boldsymbol{\delta} \in \{\mathbf{x}, \mathbf{y}\}} (\hat{n}_{\mathbf{r}\downarrow} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\uparrow})\end{aligned}$$

These two operators are “complementary”: $\sigma = \uparrow$ densities act on $\mathcal{S}_1$ for $\hat{H}_V^{\uparrow\downarrow}$, on $\mathcal{S}_2$ for $\hat{H}_V^{\downarrow\uparrow}$; $\sigma = \downarrow$ densities, the opposite. Then their sum can be expressed as:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r} \in \mathcal{L}} \sum_{\boldsymbol{\delta} \in \{\mathbf{x}, \mathbf{y}\}} (\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow})\end{aligned}\quad (2.20)$$

In the context of the HF method, Wick's theorem applied to this quartic operator gives:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} \rangle &= \langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle \\ &= \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}}\end{aligned}\quad (2.21)$$

This equality is to be intended limitedly to the subspace of states that we are considering for the HF variational search.

For the same-spin sector we define

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} \sum_{\sigma \in \{\uparrow, \downarrow\}} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma})\end{aligned}\quad (2.22)$$

Note that, with respect to the o.s. sector, we are summing on half the sites two times: one for spin $\uparrow$, one for spin $\downarrow$. Applying Wick's theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} \rangle &= \langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle \\ &= \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \rangle \langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma} \rangle \langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma}^\dagger \rangle \langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle}_{\text{Cooper}}\end{aligned}\quad (2.23)$$

As we did for $\hat{H}_U$, we analyse each separately.

Hartree terms. As it was for $\hat{H}_U$, Hartree EVs of Eqns. (2.20), (2.22) are not subject to selection rules for any symmetry of Tab. 2.1. Thus they are allowed in both sectors.

Fock terms. The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}\left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z \right] &= \left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} (\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow}) \right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} [\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \hat{c}_{\mathbf{r}\sigma}^\dagger [\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \right) \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\end{aligned}\quad (2.24)$$

where in last passage we used Eq. (2.17) and its conjugate version,

$$[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}] = \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}$$

Now, if $\sigma = \sigma'$ the last passage of Eq. (2.24) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to U^z(1). [Commutation with $\hat{S}^{+?}$]

Cooper terms. Cooper terms are not gauge invariant. The proof is analogous to the one given for $\hat{H}_U$. Instead, the o.s. and s.s. spin sectors behave differently under global spin rotations. With an identical procedure as for Eq. (2.24), we get:

$$\begin{aligned}\left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}^\dagger, \hat{S}^z \right] &= \left[\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}^\dagger, \frac{1}{2} \sum_{\mathbf{r}' \in \mathcal{L}} (\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow}) \right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} [\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}^\dagger + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \hat{c}_{\mathbf{r}\sigma}^\dagger [\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}^\dagger, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} - \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2} \right) \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}^\dagger\end{aligned}\quad (2.25)$$

We have a mirrored situation with respect to the Fock terms. For the Cooper channel, the s.s. EVs are zero if U^z(1) is unbroken, while the o.s. sector are not.

In Tab. 2.3 we summarise these considerations, listing a set of selection rules about the EVs of Eqns. (2.21), (2.23), dividing in the two sectors “o.s.” and “s.s.”. By identifying these selection rules, we set grounds for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM. We now move to the last part of this chapter: reciprocal space description of the EHM.

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	Prohibited

Table 2.3 | Summary of selection rules for the Wick's decomposition of Eqns. (2.21), (2.23) of the non-local attraction $\hat{H}_V$. An empty entry indicates no selection rule.

2.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. It is thus convenient to move to reciprocal space, by Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{-i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \quad (2.26)$$

Throughout the text, $\stackrel{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. “BZ” is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{i\mathbf{k} \cdot \mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \quad (2.27)$$

On one sublattice:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i\mathbf{k} \cdot \mathbf{r}} = \frac{1}{2} \delta_{\mathbf{k}=\mathbf{0}} \quad (2.28)$$

The notation $\mathcal{L}/2$ is shorthand to indicate $\mathcal{S}_1$ or $\mathcal{S}_2$.

Kinetic part. The kinetic operators transforms as follows:

$$\hat{H}_t \stackrel{\mathcal{F}}{=} \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \quad (2.29)$$

where we defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t (\cos k_x + \cos k_y) \quad (2.30)$$

Let us prove this first transform.

Proof 2.5 ($\mathcal{F}$ -transform of $\hat{H}_t$).

Consider the kinetic operator $\hat{H}_t$:

$$\hat{H}_t = -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{c}_{\mathbf{r}\sigma}^{\dagger} \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}]$$

Applying the convention of Eq. (2.26), we get:

$$\begin{aligned} \hat{H}_t &\stackrel{\mathcal{F}}{=} -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left(\frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \right) \left(\frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} \right) + \text{h.c.} \\ &= -2t \sum_{\mathbf{k}, \mathbf{k}'} \sum_{\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{r}} \times \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} e^{i\mathbf{k}' \cdot \delta} \end{aligned}$$

We used Eq. (2.19) and recognised that the h.c. term gives an identical contribution. We recognise in the central term the relation of Eq. (2.28). Hence,

$$\begin{aligned}\hat{H}_t &\stackrel{\mathcal{F}}{=} -2t \sum_{\mathbf{k}, \mathbf{k}'} \sum_{\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x} + e^{-ik_x}}{2} + \frac{e^{ik_y} + e^{-ik_y}}{2} \right] \\ &= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma}\end{aligned}$$

which proves Eq. (2.29).

Local repulsion. The local repulsion transforms as follows:

$$\hat{H}_U \stackrel{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \quad (2.31)$$

where sums over the variables $\mathbf{K}, \mathbf{k}, \mathbf{k}'$ must be intended as over the BZ.

Proof 2.6 ($\mathcal{F}$ -transform of $\hat{H}_U$).

Using the convention of Eq. (2.26), we get for $\hat{H}_U$:

$$\begin{aligned}\hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^{\dagger} \hat{c}_{\mathbf{r}\downarrow}^{\dagger} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \\ &\stackrel{\mathcal{F}}{=} U \sum_{\mathbf{r} \in \mathcal{L}} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i\mathbf{k}_1 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_1\uparrow}^{\dagger} e^{i\mathbf{k}_2 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_2\downarrow}^{\dagger} e^{-i\mathbf{k}_3 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4 \cdot \mathbf{r}} \hat{c}_{\mathbf{k}_4\uparrow} \\ &= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \hat{c}_{\mathbf{k}_1\uparrow}^{\dagger} \hat{c}_{\mathbf{k}_2\downarrow}^{\dagger} \hat{c}_{\mathbf{k}_3\downarrow} \hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)] \cdot \mathbf{r}} \\ &= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \hat{c}_{\mathbf{k}_1\uparrow}^{\dagger} \hat{c}_{\mathbf{k}_2\downarrow}^{\dagger} \hat{c}_{\mathbf{k}_3\downarrow} \hat{c}_{\mathbf{k}_4\uparrow} \delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}\end{aligned}$$

where we used Eq. (2.27). Let $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4 = \mathbf{K}$, and define $\mathbf{k}, \mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K} + \mathbf{k} \quad \mathbf{k}_2 \equiv \mathbf{K} - \mathbf{k} \quad \mathbf{k}_3 \equiv \mathbf{K} - \mathbf{k}' \quad \mathbf{k}_4 \equiv \mathbf{K} + \mathbf{k}'$$

Substituting, we get the expression of Eq. (2.31).

Non-local attraction. In o.s. sector, the non-local attraction transforms as:

$$\hat{H}_V^{(\text{o.s.})} \stackrel{\mathcal{F}}{=} -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \quad (2.32)$$

In the s.s. sector:

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\stackrel{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \\ &\quad \times \left[\hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}_{\mathbf{K}+\mathbf{k}\downarrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow} \right] \quad (2.33)\end{aligned}$$

Sums over the variables $\mathbf{K}, \mathbf{k}, \mathbf{k}'$ must be intended as over the BZ.

Proof 2.7 ($\mathcal{F}$ -transform of $\hat{H}_V$).

To prove Eqns. (2.32), (2.33), it is useful to start by defining:

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\delta \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}+\delta\sigma'} + \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}-\delta\sigma'})$$

with:

$$\hat{H}_V = \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\sigma, \sigma'} h_{\mathbf{r}\sigma\sigma'}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r} \pm \delta \sigma'} &= \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r} \pm \delta \sigma'}^\dagger \hat{c}_{\mathbf{r} \pm \delta \sigma'} \hat{c}_{\mathbf{r}\sigma} \\ &\stackrel{\mathcal{F}}{=} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} e^{\pm i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} \hat{c}_{\mathbf{k}_1 \sigma}^\dagger \hat{c}_{\mathbf{k}_2 \sigma'}^\dagger \hat{c}_{\mathbf{k}_3 \sigma'} \hat{c}_{\mathbf{k}_4 \sigma} \end{aligned}$$

Then:

$$\begin{aligned} h_{\mathbf{r}\sigma\sigma'} &\stackrel{\mathcal{F}}{=} -\frac{V}{(L_x L_y)^2} \sum_{\delta \in \{\underline{x}, \underline{y}\}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} \\ &\quad \times \left(e^{i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} + e^{-i(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta} \right) \hat{c}_{\mathbf{k}_1 \sigma}^\dagger \hat{c}_{\mathbf{k}_2 \sigma'}^\dagger \hat{c}_{\mathbf{k}_3 \sigma'} \hat{c}_{\mathbf{k}_4 \sigma} \\ &= -\frac{2V}{(L_x L_y)^2} \sum_{\delta \in \{\underline{x}, \underline{y}\}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} \cos[(\mathbf{k}_2 - \mathbf{k}_3) \cdot \delta] \hat{c}_{\mathbf{k}_1 \sigma}^\dagger \hat{c}_{\mathbf{k}_2 \sigma'}^\dagger \hat{c}_{\mathbf{k}_3 \sigma'} \hat{c}_{\mathbf{k}_4 \sigma} \end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (2.28) gives back momentum conservation

$$\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}/2} e^{i[(\mathbf{k}_1 + \mathbf{k}_2) - (\mathbf{k}_3 + \mathbf{k}_4)] \cdot \mathbf{r}} = \frac{1}{2} \delta_{\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4}$$

where we used Eq. (2.28). Define $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$ as we did for the local repulsion, and

$$\delta \mathbf{k} \equiv \mathbf{k} - \mathbf{k}'$$

Hence:

$$\begin{aligned} \hat{H}_V &= \sum_{\mathbf{r} \in \mathcal{L}/2} \sum_{\sigma, \sigma'} h_{\mathbf{r}\sigma\sigma'} \\ &\stackrel{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\sigma \sigma'} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \cos(\delta \mathbf{k} \cdot \delta) \hat{c}_{\mathbf{K} + \mathbf{k} \sigma}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k} \sigma'}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k}' \sigma'} \hat{c}_{\mathbf{K} + \mathbf{k}' \sigma} \\ &= -\frac{V}{L_x L_y} \sum_{\sigma \sigma'} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K} + \mathbf{k} \sigma}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k} \sigma'}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k}' \sigma'} \hat{c}_{\mathbf{K} + \mathbf{k}' \sigma} \end{aligned} \quad (2.34)$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$\begin{aligned} \hat{H}_V &\stackrel{\mathcal{F}}{=} \overbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K} + \mathbf{k} \sigma}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k} \sigma}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k}' \sigma} \hat{c}_{\mathbf{K} + \mathbf{k}' \sigma}}^{\text{s.s.}} \\ &\quad \underbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \hat{c}_{\mathbf{K} + \mathbf{k} \sigma}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k} \bar{\sigma}}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k}' \bar{\sigma}} \hat{c}_{\mathbf{K} + \mathbf{k}' \sigma}}_{\text{o.s.}} \end{aligned}$$

The s.s. part confirms Eq. (2.33). The o.s. part can be simplified: since

$$\hat{c}_{\mathbf{K} + \mathbf{k} \sigma}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k} \bar{\sigma}}^\dagger \hat{c}_{\mathbf{K} - \mathbf{k}' \bar{\sigma}} \hat{c}_{\mathbf{K} + \mathbf{k}' \sigma} = \hat{c}_{\mathbf{K} - \mathbf{k} \bar{\sigma}}^\dagger \hat{c}_{\mathbf{K} + \mathbf{k} \sigma}^\dagger \hat{c}_{\mathbf{K} + \mathbf{k}' \sigma} \hat{c}_{\mathbf{K} - \mathbf{k}' \bar{\sigma}}$$

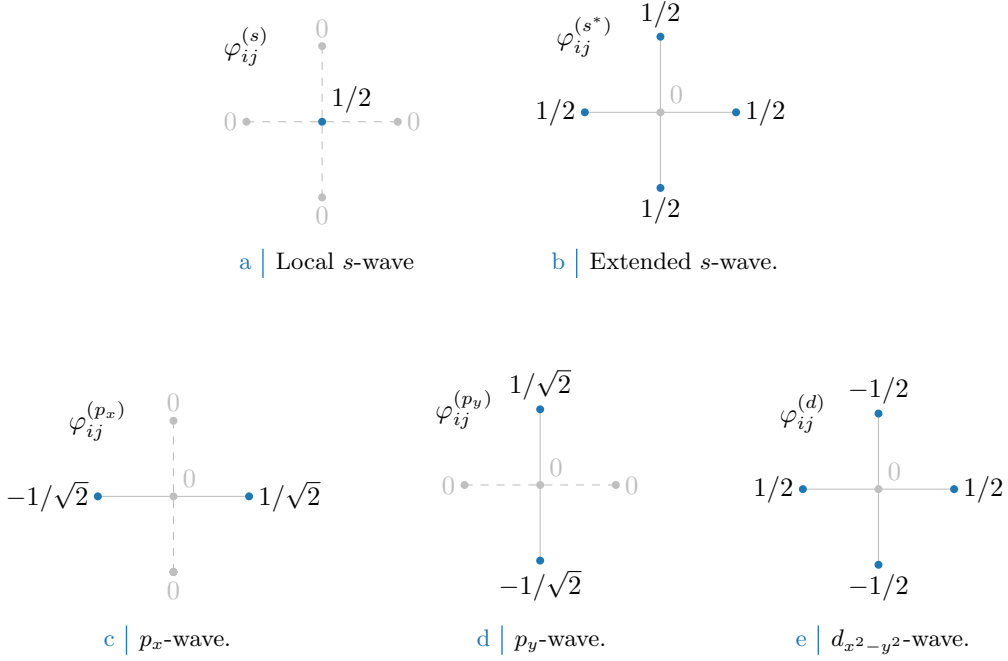


Figure 2.4 Form factors at different topologies, as listed in Tab. 2.4. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for φ_{δ} , while dashed lines represent vanishing factors.

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \rightarrow -\mathbf{k} \quad \mathbf{k}' \rightarrow -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part. This confirms Eq. (2.32)

All these expressions are going to be used in the analysis of each phase. In reciprocal space, following the general decomposition presented in Eq. (??) for a quartic operator $\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$, we distinguish contractions as follows:

- Hartree contractions:

$$\overline{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}} \quad \hat{c}_{\mathbf{k}_1}^\dagger \overline{\hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}$$

- Fock contractions:

$$\overline{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4} \quad \hat{c}_{\mathbf{k}_1}^\dagger \overline{\hat{c}_{\mathbf{k}_2}^\dagger \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}$$

- Cooper contractions:

$$\overline{\hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4} \quad \hat{c}_{\mathbf{k}_1}^\dagger \hat{c}_{\mathbf{k}_2}^\dagger \overline{\hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}$$

with appropriate sign factors. With the EHM expressed in reciprocal space, there is no general feature left to address. We can move to the detailed discussion of each phase.

2.4 Square lattice spatial symmetry channels

This short conclusive section is apparently rather uncorrelated with all we said in the previous parts. We here introduce a set of orthonormal functions that results particularly useful to represent quantities defined on NNs sites on the square lattice. Let us start from a generic function of two variables $f(\mathbf{r}, \mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself ($\mathbf{r}' = \mathbf{r}$) and its NNs $\mathbf{r}' = \mathbf{r} \pm \boldsymbol{\delta}$, with $\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$. In other words,

$$f(\mathbf{r}, \mathbf{r}') \neq 0 \quad \Longleftrightarrow \quad |\mathbf{r} - \mathbf{r}'| = 0, 1$$

Structure	Form factor	Graph
s -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(s)} = \delta_{\mathbf{r}\mathbf{r}'}$	Fig. 2.4a
Extended s -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} = \frac{1}{2} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}]$	Fig. 2.4b
p_x -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} = \frac{1}{\sqrt{2}} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}}]$	Fig. 2.4c
p_y -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} = \frac{1}{\sqrt{2}} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}]$	Fig. 2.4d
$d_{x^2-y^2}$ -wave	$\varphi_{\mathbf{r}\mathbf{r}'}^{(d)} = \frac{1}{2} [\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}]$	Fig. 2.4e

Table 2.4 | Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 2.4. Subscript $x^2 - y^2$ is omitted for notational clarity.

in units of the lattice step. The following identity holds:

$$\begin{aligned} f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}') \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} \\ &= f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} + \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} + \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} \right) \end{aligned}$$

where we defined the set of orthonormal functions $\varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}$ listed explicitly in Tab. 2.4 and sketched in Fig. 2.4. The orthonormality condition reads:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\mathbf{r}'} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma')*} = \delta_{\gamma\gamma'}$$

Similarly, also the following relations hold:

$$\begin{aligned} f(\mathbf{r}, \mathbf{r}) &= f(\mathbf{r}, \mathbf{r}') \varphi_{\mathbf{r}\mathbf{r}'}^{(s)} \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} + \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} - \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} \right) \\ f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} - \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} + \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} \right) \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} - \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} - \sqrt{2} \varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} \right) \end{aligned}$$

Thus, defining:

$$\begin{aligned} f^{(s)} &\equiv f(\mathbf{r}, \mathbf{r}) \\ f^{(s^*)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) \\ f^{(p_x)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) \\ f^{(p_y)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) \\ f^{(d)} &\equiv f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) - f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) - f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) \end{aligned}$$

we get the decomposition:

$$f(\mathbf{r}, \mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}$$

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$ -transformed functions of Tab. 2.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma')*} = \delta_{\gamma\gamma'}$$

Structure	Structure factor	Graph
s -wave	$\varphi_{\mathbf{k}}^{(s)} = 1$	Fig. 2.4a
Extended s -wave	$\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$	Fig. 2.4b
p_x -wave	$\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2} \sin k_x$	Fig. 2.4c
p_y -wave	$\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2} \sin k_y$	Fig. 2.4d
$d_{x^2-y^2}$ -wave	$\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$	Fig. 2.4e

Table 2.5 | Structure factors derived from the correlation structures of Tab. 2.4. The functions hereby defined are orthonormal, and are plotted in Fig. 2.5.

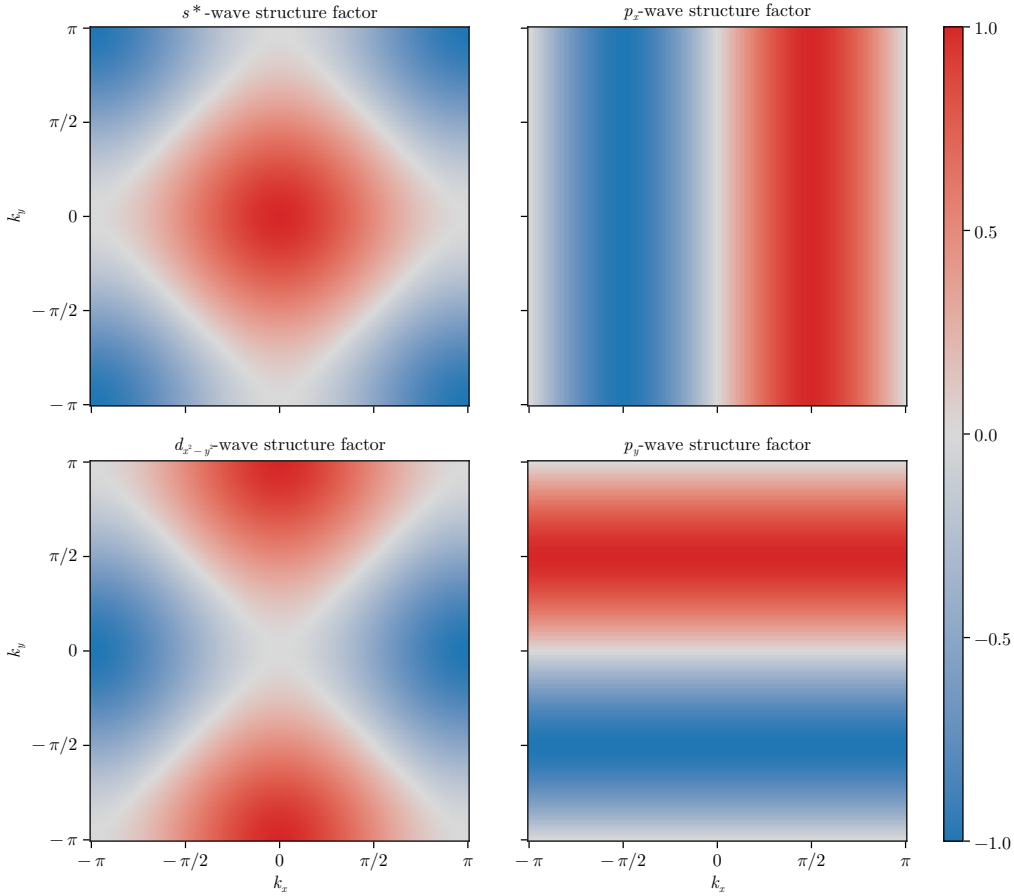


Figure 2.5 | Representation of the various structure factors listed in Tab. 2.5 on the BZ. For the s^* and $d_{x^2-y^2}$ -wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

The functions of Tab. 2.5 are plotted in the BZ in Fig. 2.5.

We will refer to these harmonics as “symmetry channels”, indicating the various spatial components of the order parameter associated to a given phase transition. As we will see, indeed, we consider order parameters that have specific transformation properties under planar rotations. To distinguish the various spatial harmonics of the order parameter is useful in order to apply symmetry-based selection rules.

Finally, a very useful mathematical identity is given by the following argument. Consider Eq. (2.34), where is present the factor $\cos(\delta k_x) + \cos(\delta k_y)$ (with $\delta k_x = k_x - k'_x$ and $\delta k_y = k_y - k'_y$).

Expanding,

$$\cos(\delta k_x) + \cos(\delta k_y) = c_x c'_x + c_y c'_y + s_x s'_x + s_y s'_y$$

In last passage, for the sake of readability, we used the notations

$$c_\ell \equiv \cos k_\ell \quad s_\ell \equiv \sin k_\ell \quad c'_\ell \equiv \cos k'_\ell \quad s'_\ell \equiv \sin k'_\ell$$

Group the four terms,

$$\underbrace{(c_x c'_x + c_y c'_y)}_{\text{Symmetric}} + \underbrace{(s_x s'_x + s_y s'_y)}_{\text{Anti-symmetric}} \quad (2.35)$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \rightarrow (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}, \mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned} \cos(\delta k_x) + \cos(\delta k_y) &= \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\ &+ s_x s'_x && (p_x\text{-wave}) \\ &+ s_y s'_y && (p_y\text{-wave}) \\ &+ \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave}) \end{aligned}$$

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2} \sum_{\gamma} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \quad (2.36)$$

$$= \frac{1}{2} \sum_{\gamma} \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \quad (2.37)$$

This decomposition will result particularly handy in later calculations.

Chapter 3

The normal phase

In this chapter we apply the HF methods sketched in Sec. ?? to the “normal phase”. We are studying the model in a state where no symmetry of those enumerated in Tab. 2.1 is broken. Even though this phase may appear not so interesting, it actually shows a significant physical difference with respect to the conventional perfectly degenerate Fermi gas.

HF methods applied in this context will lead us to a self-consistent hopping renormalisation effect. For certain values of the non-local attraction V and the doping level, a complete Mott localisation effect takes place. The resulting bands shrinking and flattening is the main net effect of the presence of $\hat{H}_V$ in the EHM.

This chapter is divided in three main parts. First, we discuss the symmetries of the normal phase, following the general discussion of Chap. 2. Second, we apply HF methods to derive the self-consistent approximated hamiltonian and impose some theoretical constraints on the self-consistent solution. Lastly, we present numeric results.

3.1 The normal phase and its symmetries

In the normal phase no symmetry is broken. Thus, the HF wavefunction approximating the ground state must exhibit the full $U(2) \otimes C_4 \otimes T_1^{(x)} \otimes T_1^{(y)}$ symmetry discussed in Sec. 2.1. We take a slight detour from this, and instead allow for

$$C_4 \xrightarrow{\text{SSB}} C_2$$

This is the only potential SSB we consider. It will turn out in Sec. 3.4.2 that even allowing for breaking of rotational symmetry, the ground-state actually does not break C_4 . This can be seen as a convolute way to confirm that there is not a normal-nematic phase. While this may seem as an useless complication, the derivation will set grounds for actual nematic effects found in the superconducting phase.

Our HF method relies on the application of Wick’s theorem on the quartic part of the interacting hamiltonian, expressed in Eq. (??). We will decompose the quartic operators according to Eqns. (??), (??). Recalling Tabs. 2.2 and 2.3, for the normal phase we must consider the following contributes to the EVs of Eqns. (2.16), (2.21), (2.23):

$\hat{H}_U$	Hartree contractions
$\hat{H}_V$ (o.s. sector)	Hartree contractions
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

As we will see in detail and is summarised in Tab. 3.1, Hartree contractions give a chemical potential shift, while the unique Fock contraction effectively renormalises the bare bands.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \bar{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree Fock	μ shift by nzV Bands renormalisation and resymmetrization

Table 3.1 | Relevant Wick's contractions and net effect in the normal phase.

3.2 Application of HF methods to the normal phase

The quartic part of $\hat{H}$ is given by the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. Applying HF methods to them, we get the decomposition of Eqns. (??), (??). In the next section we analyse the Hartree channel, which contains all Hartree-like terms. We will see that $\hat{H}_U$ and $\hat{H}_V$ are approximated, in this channel, by a multiple of the number operator. In the section that follows, we deal with Fock terms, and show how $\hat{H}_V$ is approximated, in that channel, by a kinetic-like operator.

3.2.1 Hartree effects: chemical potential renormalisation

Both the two-body operators $\hat{H}_U$, $\hat{H}_V$ have non-zero Hartree EVs. We deal with them separately, and show that both lead to a renormalisation of μ .

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as specified in Eq. (??)

$$\hat{H}_U \stackrel{\text{HF}}{\simeq} U \sum_i [\langle \hat{n}_{i\uparrow} \rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow} \langle \hat{n}_{i\downarrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle \langle \hat{n}_{i\uparrow} \rangle]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma} \rangle = n \quad (3.1)$$

which gives

$$\begin{aligned} \hat{H}_U &\stackrel{\text{HF}}{\simeq} nU \sum_i [\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}] - U \sum_i n^2 \\ &= nU \times \hat{N} - E_{\text{H}/U}^{(\text{N})} \end{aligned} \quad (3.2)$$

being $\hat{N}$ the total number operator and $E_{\text{H}/U}^{(\text{N})}$ the “local Hartree contraction energy” (also known as double counting term),

$$E_{\text{H}/U}^{(\text{N})} = L_x L_y \times n^2 U \quad (3.3)$$

This energy contributes negatively to the ground-state energy, for $U > 0$. The chemical potential is corrected by

$$\mu \rightarrow \mu - nU \quad (3.4)$$

The non-local attraction further corrects μ .

Non-local attraction. The non-local attraction is divided in two sectors, as in Eq. (2.18). From Eq. (??), its Hartree-like terms are:

$$\begin{aligned} \hat{H}_V &\stackrel{\text{HF}}{\simeq} \overbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle]}^{\text{s.s.}} \\ &\quad \underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle]}_{\text{o.s.}} + (\text{all the rest}) \end{aligned}$$

where “all the rest” collects all non-Hartree terms. Recalling the Ansatz of Eq. (3.1), we get

$$\hat{H}_V \stackrel{\text{HF}}{\simeq} \underbrace{-nV \sum_{\langle ij \rangle} \sum_{\sigma} [\hat{n}_{j\sigma} + \hat{n}_{i\sigma}]}_{\text{s.s.}} \underbrace{-nV \sum_{\langle ij \rangle} \sum_{\sigma} [\hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma}]}_{\text{o.s.}} - E_{\text{H}/V}^{(\text{N})} + (\text{all the rest}) \quad (3.5)$$

with $E_{\text{H}/V}^{(\text{N})}$ the normal state shift to energy brought by $\hat{H}_V$:

$$\begin{aligned} E_{\text{H}/V}^{(\text{N})} &= -V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle + \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle] \\ &= -2V \sum_{\langle ij \rangle} \sum_{\sigma} n^2 \\ &= -L_x L_y \times 2zn^2 V \end{aligned} \quad (3.6)$$

Note the similarity with Eq. (??).

Getting back to Eq. (3.5), both sums give a number operator. Note that we sum over NNs: each site is connected to $z = 4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E_{\text{H}/V}^{(\text{N})} + (\text{all the rest}) \quad (3.7)$$

This accounts for the final Hartree shift of μ ,

$$\mu \rightarrow \mu + 2znV \quad (3.8)$$

Collecting everything, we get the HF approximation to the two interactions in the Hartree channel:

$$\hat{H}_U + \hat{H}_V \stackrel{\text{HF}}{\simeq} -n(2zV - U)\hat{N}$$

Considering the net shift to μ due to both interactions by combining Eqns. (3.4), (3.8), we get the final RMP anticipated in Tab. 3.1,

$$\tilde{\mu} \equiv \mu + n(2zV - U) \quad (3.9)$$

This result will remain valid throughout all phases discussed in this text.

3.2.2 Fock effects: bare bands renormalisation

We move to Fock terms. From Wick’s decomposition of $\hat{H}_V$, the unique allowed Fock term comes from the same-spin sector. This selection rules is implied by the SU(2) symmetry of the normal phase and was stated in Tab. 2.3. Then, from Eq. (??):

$$\hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \right] \quad (3.10)$$

Note the plus sign in front of the displayed term: it comes from the minus sign in Fock term of Eq. (??).

Consider the Fourier Transform $\mathcal{F}$ given in Eq. (2.33), which gives

$$\begin{aligned} \hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) \\ + \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E_{\text{F}/V}^{(\text{N})} \end{aligned} \quad (3.11)$$

Here, the 2 prefactor comes from the fact that the first two operators in square brackets in Eq. (3.10) are equal in reciprocal space. The double counting energy shift is given by

$$E_{\text{F}/V}^{(\text{N})} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \quad (3.12)$$

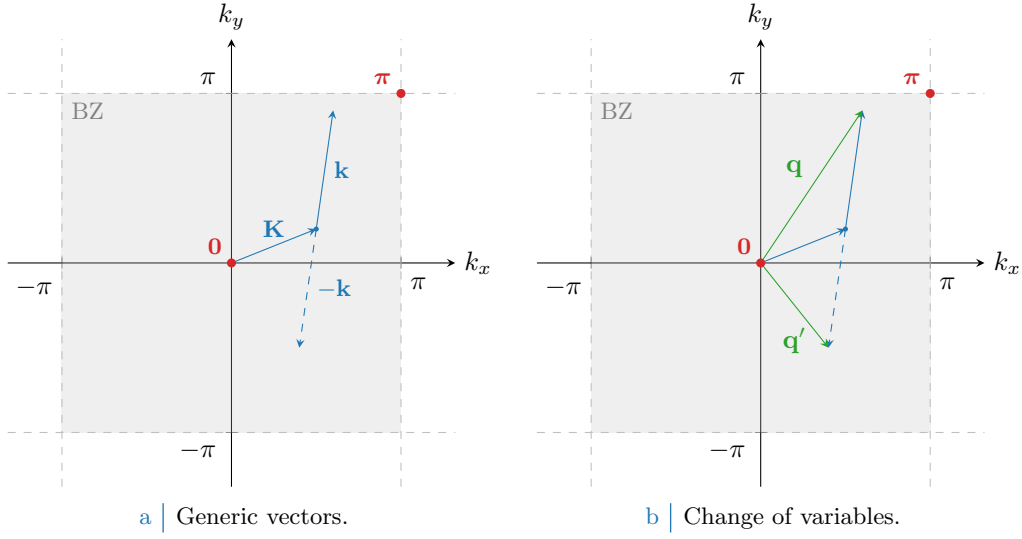


Figure 3.1 | Representation of the variable change employed to pass from Eq. (3.17) to (3.18). In Fig. 3.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in \text{BZ}$. In Fig. 3.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

We define

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \quad u_{\mathbf{k}\sigma} = u_{\mathbf{k}\sigma}^* \quad (3.13)$$

Using the functions of Tab. 2.5, we define its harmonics:

$$u_{\sigma}^{(\gamma)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \quad (3.14)$$

The non-local attraction $\hat{H}_V$ is approximated, in Fock channel, by:

$$\hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \left[u_{\sigma}^{(s^*)} \varphi_{\mathbf{k}}^{(s^*)*} + u_{\sigma}^{(d)} \varphi_{\mathbf{k}}^{(d)*} \right] \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} - E_{\text{F}/V}^{(\text{N})} \quad (3.15)$$

Let us prove these this.

Proof 3.1 (Proof of Eq. (3.15)).

The normal phase is described by a perfectly degenerate Fermi gas, hence it must hold:

$$\langle \hat{c}_{\mathbf{k}_1\sigma_1}^\dagger \hat{c}_{\mathbf{k}_2\sigma_2} \rangle = \delta_{\mathbf{k}_1=\mathbf{k}_2} \delta_{\sigma_1=\sigma_2} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \quad (3.16)$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

Consider now Eq. (3.11). Applying Eq. (3.16), it becomes:

$$\begin{aligned} \hat{H}_V \stackrel{\text{HF}}{\simeq} (\text{Hartree channel}) \\ + \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E_{\text{F}/V}^{(\text{N})} \end{aligned} \quad (3.17)$$

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 3.1.

Being (for $\ell = x, y$)

$$\begin{aligned}\delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell\end{aligned}$$

we reduce Eq. (3.11) to

$$\begin{aligned}\hat{H}_V^{\text{HF}} \text{ (Hartree channel)} \\ + \frac{2V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \underbrace{\langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} \rangle}_{u_{\mathbf{q}\sigma}} \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} - E_{\text{F}/V}^{(\text{N})}\end{aligned} \quad (3.18)$$

where we recognised $u_{\mathbf{q}\sigma}$, as was defined in Eq. (3.13). Using Eq. (2.36), we obtain

$$\begin{aligned}\hat{H}_V^{\text{HF}} \text{ (Hartree channel)} \\ + \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \times V \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} - E_{\text{F}/V}^{(\text{N})}\end{aligned} \quad (3.19)$$

where we used the functions defined in Tab. 2.5. We recognise the harmonics $u_{\sigma}^{(\gamma)}$ from Eq. (3.14),

$$\hat{H}_V^{\text{HF}} \text{ (Hartree channel)} + V \sum_{\mathbf{q}' \in \text{BZ}} \sum_{\gamma, \sigma} u_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} - E_{\text{F}/V}^{(\text{N})} \quad (3.20)$$

which is almost Eq. (3.15).

To complete the proof, we recognise that in the normal phase inversion symmetry is not broken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

This follows from the fact that in the normal phase the EV of the operator $\hat{n}_{\mathbf{k}\sigma}$ must be inversion-symmetric. It follows that antisymmetric harmonics are zero,

$$u_{\sigma}^{(p_\ell)} = 0 \quad \text{for } \ell \in \{x, y\}$$

Using this observation in Eq. (3.20), we obtain Eq. (3.15).

Composing Eq. (3.15) with Eqns. (3.2), (3.7) we obtain the quadratic HF approximation to the EHM, for the normal phase. Its parameters are:

1. The chemical potential $\tilde{\mu}$, which is fixed by density and determined during computations;
2. The two HFPs $u_{\sigma}^{(s^*)}$, $u_{\sigma}^{(d)}$.

To find the ground-state, we must determine self-consistently these two HFPs through the self-consistency Eq. (3.14).

The bare bands $\epsilon_{\mathbf{k}}$ are s^* -wave symmetric,

$$\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y) = -4t\varphi_{\mathbf{k}}^{(s^*)}$$

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_{\sigma}^{(s^*)} V(\cos k_x + \cos k_y) + u_{\sigma}^{(d)} V(\cos k_x - \cos k_y) \quad (3.21)$$

This equation defines the bands renormalisation anticipated in Tab. 3.1, and is actually spin-independent. The fact that we have both s^* and a d -wave components introduces an anisotropy that is discussed in next section. However, we anticipate here that, as will result from Sec. 3.4.2, the d -wave harmonic will be suppressed. For now we keep this component ignoring these anticipations.

Symmetry	Self-consistency equation	Graph
s^* -wave	$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 2.4b
$d_{x^2-y^2}$ -wave	$u^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 2.4e

Table 3.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

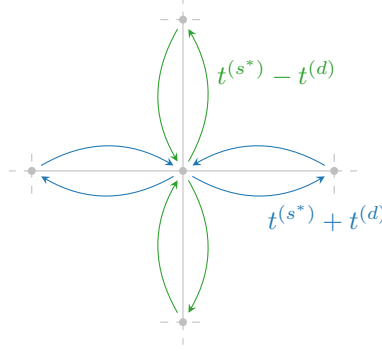


Figure 3.2 | Graphic rendition of anisotropic hopping, with a s^* -wave component (symmetric by $\pi/2$ rotations) and a d -wave component (antisymmetric).

while the contraction energy is given by:

$$E_{\text{F}/V}^{(\text{N})} = L_x L_y \times \frac{V}{2} \sum_{\sigma} \left[|u_{\sigma}^{(s^*)}|^2 + |u_{\sigma}^{(d)}|^2 \right] \quad (3.22)$$

Finally, let us derive the Fock contraction energy by expanding Eq. (3.12):

$$\begin{aligned}
E_{\text{F}/V}^{(\text{N})} &= \frac{V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}_{\mathbf{q}\sigma}^{\dagger} \hat{c}_{\mathbf{q}'\sigma} \rangle \langle \hat{c}_{\mathbf{q}\sigma}^{\dagger} \hat{c}_{\mathbf{q}'\sigma} \rangle \\
&= L_x L_y \times \frac{V}{2} \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right] \\
&= L_x L_y \times \frac{V}{2} \sum_{\sigma} \left[|u_{\sigma}^{(s^*)}|^2 + |u_{\sigma}^{(d)}|^2 \right] \quad (3.23)
\end{aligned}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The squared spectral amplitude of $u_{\mathbf{k}\sigma}$ here is summed, and contributes to contraction energy linearly in V . This suggests that a mixed-symmetry band could be energetically preferred.

3.2.3 Nematicity: extending hopping to d -wave bands

For the normal phase, there is no reason to expect nematic effects, nor to exclude them. They are observed in superconducting phase [23]. Thus, we allow the possibility.

From Tabs. 2.4 and 2.5 we can inverse-Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma \in \{s^*, d\}} t_{\sigma}^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_{\sigma}^{(s^*)} = t - \frac{u_{\sigma}^{(s^*)} V}{2} \quad t_{\sigma}^{(d)} = -\frac{u_{\sigma}^{(d)} V}{2}$$

This is what we intend for “nematic effects”. The possibility that planar rotational symmetry is broken in such a way that the kinetic operators in x and y directions differ. A schematics of anisotropic hopping is given in Fig. 3.2: for each bond linking two sites, the hopping parameter can be a positive or negative combination of the two components, whether the link direction is vertical or horizontal. The solution needs to be invariant under exchange $x \leftrightarrow y$, thus the actual SSB is $C_4 \xrightarrow{\text{SSB}} C_2/\mathbb{Z}_2$.

The fact that hopping is here anisotropic does not break full translational invariance $T_1^{(x)} \otimes T_1^{(y)}$. Imposing a free degenerate Fermi gas ground state in Eq. (3.16) leads to a uniform density. This is proven by Fourier-transforming the site number operator according to the convention of Eq. (2.26),

$$\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}') \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma}$$

Then the on-site density EV is

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry. The fact that hopping is allowed to be anisotropic does not interfere with that.

Note that for null HFPs,

$$u_{\sigma}^{(s^*)} = 0 \quad u_{\sigma}^{(d)} = 0$$

we have, recalling Tab. 2.4:

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{t}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right]$$

The fact that $t/2$ appears here instead of t is a consequence of the equivalence:

$$-t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.} \right] = -\frac{t}{2} \sum_{i,j \in \mathcal{S}} \sum_{\sigma} \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.} \right]$$

where we including a prefactor that corrects double-counting. The RMP $\tilde{t}_{ij\sigma}$ must be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2} \sum_{i,j \in \mathcal{S}} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.} \right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

Recall the functions of Tab. 2.4. We have the identities:

$$\begin{aligned} \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} &= \varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} + \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} \\ \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} &= \varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} - \varphi_{\mathbf{r}\mathbf{r}'}^{(d)} \end{aligned}$$

Then we can define the renormalised hopping on each direction as:

$$\begin{aligned} \tilde{t}^{(x)} &\equiv t - \frac{u^{(s^*)} + u^{(d)}}{2} V \\ \tilde{t}^{(y)} &\equiv t - \frac{u^{(s^*)} - u^{(d)}}{2} V \end{aligned}$$

Both components depend linearly on V . The HFP $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V , while $u^{(d)}$ measures the anisotropy,

$$u^{(s^*)} = \frac{(t - \tilde{t}^{(x)}) + (t - \tilde{t}^{(y)})}{V} \quad (3.24)$$

$$u^{(d)} = \frac{\tilde{t}^{(x)} - \tilde{t}^{(y)}}{V} \quad (3.25)$$

It follows that if $u^{(d)} > 0$, the hopping in direction x is boosted while in direction y is damped. If $u^{(d)} < 0$, the contrary.

This discussion is general, and not specific to the normal phase. In next chapters we will interpret phase renormalisation as in this section.

3.3 Normal free energy density

In the end, we want to compare phases. Thus we need a way to extract the free energy density of the approximated ground-state for each phase. The procedure sketched here will be re-used in next chapters. Let

$$f^{(N)} \equiv \frac{F_{\text{MFT}}^{(N)}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} - \mu \hat{N} \simeq \sum_{\mathbf{k}, \sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} + E_c^{(N)} - \tilde{\mu} \hat{N}$$

where $E_c^{(N)}$ is the (negative) total contraction energy for the normal phase,

$$E_c^{(N)} \equiv - \left[E_{\text{H}/U}^{(N)} + E_{\text{H}/V}^{(N)} + E_{\text{F}/V}^{(N)} \right]$$

having included the various contraction energies from Eqns. (3.3), (3.6) and (3.23).

Let us now take together all terms. From now on, we will omit spin index σ , making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 3.1. For a free Fermi gas, the free energy density is made of four contributions:

1. The free particles ground state energy density,

$$e_g^{(N)} = \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e_c^{(N)} &\equiv \frac{E_c^{(N)}}{L_x L_y} \\ &= -n^2 U + V \left[2zn^2 - |u^{(s^*)}|^2 - |u^{(d)}|^2 \right] \end{aligned}$$

3. The entropic contribution to free energy

$$e_s^{(N)} \equiv -Ts$$

with $T = 1/k_B \beta$ and s the free Fermi gas entropy density for a single spin sector [29],

$$s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \quad (3.26)$$

We use the identity:

$$\ln \left(1 - \frac{1}{e^x + 1} \right) = x + \ln \left(\frac{1}{e^x + 1} \right)$$

and, substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$, reduce the above expression to:

$$\begin{aligned} s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} &\left[\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right. \\ &\left. - \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned}$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(N)} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(N)} = \mu n$$

Note here that we are using the “physical” value of μ , not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. ??.

Thus, composing everything

$$\begin{aligned} f^{(N)} &\equiv e_g^{(N)} + e_c^{(N)} + e_s^{(N)} + e_n^{(N)} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - V \left[|u^{(s^*)}|^2 + |u^{(d)}|^2 \right] + [\mu + n(2zV - U)] n \end{aligned}$$

In Sec. ?? we argued that our iterative scheme computes $\tilde{\mu}$, not μ . Using Eq. (3.9), we see

$$\mu + n(2zV - U) = \tilde{\mu}$$

Then, the final expression is:

$$f^{(N)} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - V \left[|u^{(s^*)}|^2 + |u^{(d)}|^2 \right] + \tilde{\mu} n \quad (3.27)$$

including all effects.

3.4 Theoretical constraints on the normal phase HFPs

[Move this section to appendix?]

Within this section we discuss some theoretical constraints on the two HFPs of the normal phase. The discussion is divided in two parts: first we consider the simpler case of unviolated C_4 symmetry, then we move to anisotropic hopping and $C_4 \xrightarrow{\text{SSB}} C_2$.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$ -wave band. Let in general for notational simplicity

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)$$

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^* -wave and d -wave structure factors already presented in Fig. 2.5, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\text{MBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\} \quad (3.28)$$

$$\text{NBZ} \equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \quad (3.29)$$

represented in the top-left panel of Fig. 3.3. The reason for the name of the MBZ will be clear in Chap. ??. In the rest of the panels of Fig. 3.3 are reported examples of mixed symmetry bands, for positive components. The respective opposite sign plots are obtained by the following rules:

$$\begin{aligned} \epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} < 0 &\implies \text{Invert colors in Fig. 3.3} \\ \epsilon^{(s^*)} > 0 \quad \text{and} \quad \epsilon^{(d)} < 0 &\implies \text{Rotate plots by } \pi/2 \text{ in Fig. 3.3} \\ \epsilon^{(s^*)} < 0 \quad \text{and} \quad \epsilon^{(d)} > 0 &\implies \text{Rotate plots by } \pi/2 \text{ and invert colors in Fig. 3.3} \end{aligned}$$

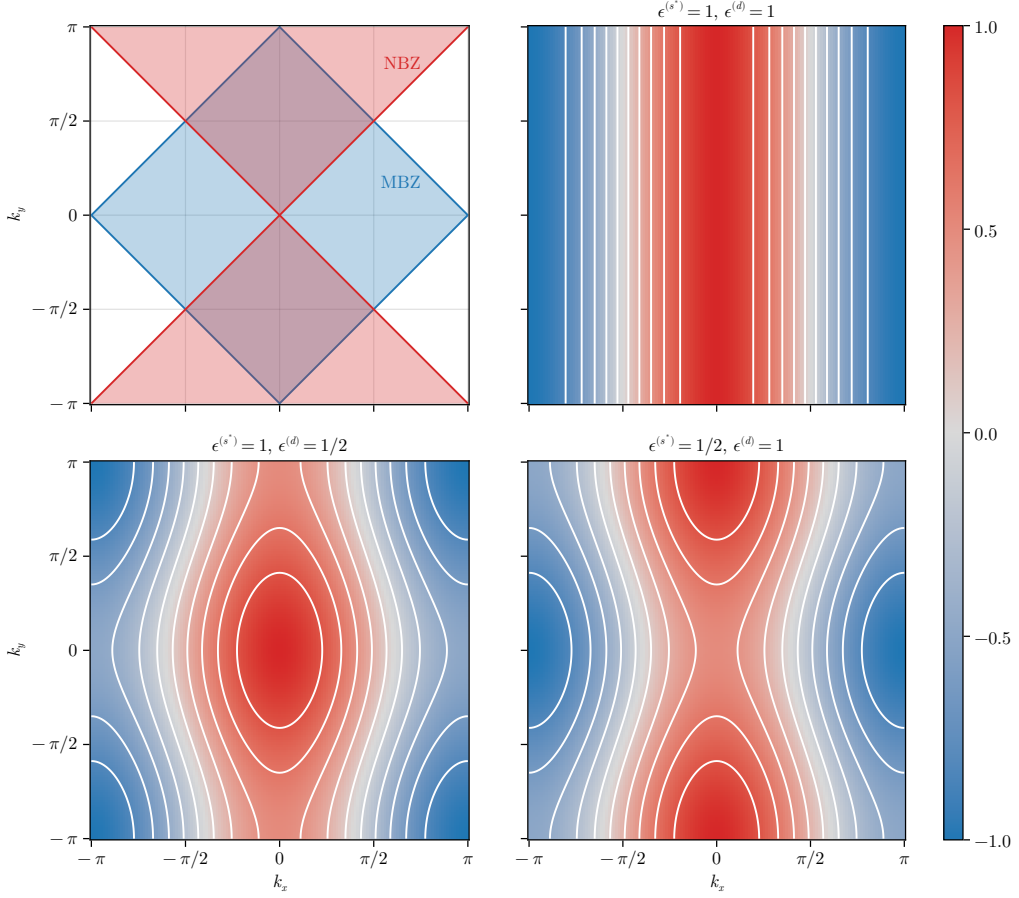


Figure 3.3 | Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\text{MBZ} = \mathcal{A} \cup \mathcal{B} \quad \text{NBZ} = \mathcal{A} \cup \mathcal{C}$$

being, accordingly to Fig. 3.4,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2 \quad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2 \quad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2$$

These domains satisfy:

$$\begin{aligned} \mathcal{A} &\equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \\ &= \text{MBZ} \cap \text{NBZ} \end{aligned} \quad (3.30)$$

$$\begin{aligned} \mathcal{B} &\equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \leq 0 \right\} \\ &= \text{MBZ} \setminus \text{NBZ} \end{aligned} \quad (3.31)$$

$$\begin{aligned} \mathcal{C} &\equiv \left\{ \mathbf{k} \in \text{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \\ &= \text{NBZ} \setminus \text{MBZ} \end{aligned} \quad (3.32)$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \quad \implies \quad \tilde{\epsilon}_{(k_x, k_y)} = 0$$

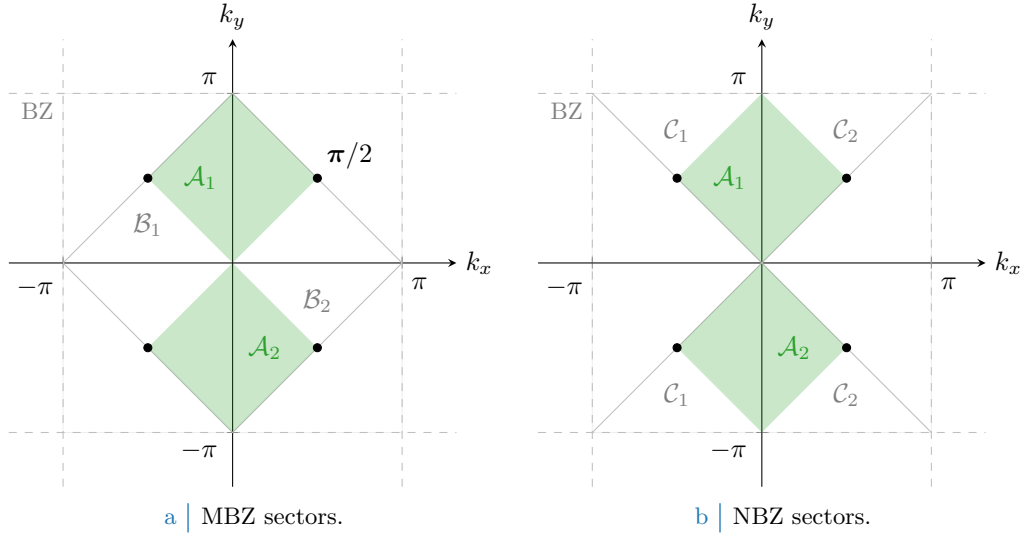


Figure 3.4 | Depiction of the sectors composing the MBZ (Fig. 3.4a) and the NBZ (Fig. 3.4b), based on the divisions shown in the top-left panel of Fig. 3.3. The full green zone composes $\mathcal{A}$, the region where both the s^* -wave and d -wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^* -wave structure factors is positive-definite while the d -wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

since only cosines are involved. This identifies four node points, represented as black dots in Fig. 3.4.

3.4.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

The following chain of deductions leads us to a constraint on the possible values of $u^{(s^*)}$:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

$$\sum_{\mathbf{k} \in \text{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k} \in \text{MBZ}} [g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}]$$

where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu})] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})] \end{aligned} \quad (3.33)$$

having we used the antisymmetry of s^* -wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \implies u^{(s^*)} \geq 0$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0$$

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})]$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

thus we have

$$u^{(s^*)} \geq 2t/V \implies u^{(s^*)} \leq 0$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V$$

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$C_4 \implies 0 < u^{(s^*)} < 2t/V \quad (3.34)$$

The effective hopping is smaller than the original hopping. We can denominate this net effect “bands shrinking”.

3.4.2 Natural suppression of nematicity within $C_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$C_4 \xrightarrow{\text{SSB}} C_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (3.21). From the following chain of deductions we conclude that C_4 is not broken:

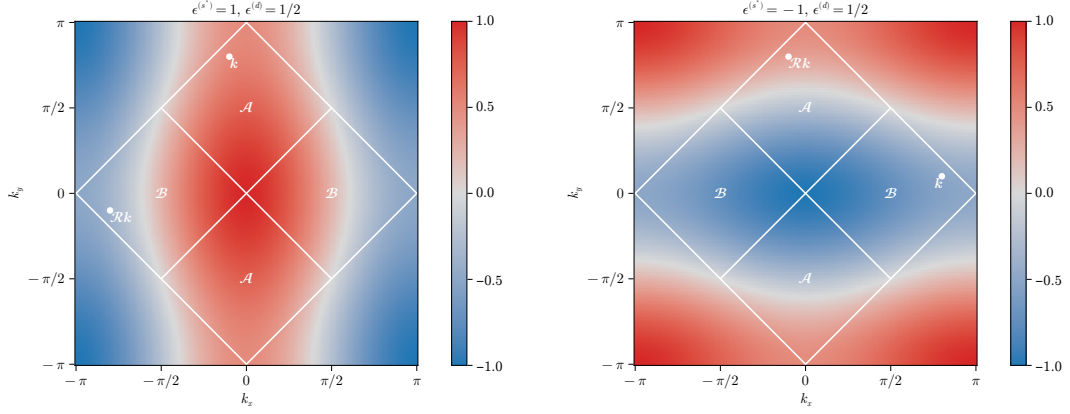
1. Let for instance

$$u^{(s^*)} \geq 2t/V > 0 \quad u^{(d)} > 0 \implies \epsilon^{(s^*)} \geq 0 \quad \epsilon^{(d)} > 0$$

In Fig. 3.5a an example band is shown for $\epsilon^{(s^*)} = 1$, $\epsilon^{(d)} = 1/2$. Note that if $u^{(s^*)} = 2t/V$, the renormalised band is purely d -wave since $\epsilon^{(s^*)} = 0$.

Recall Eq. (3.33), and separate in two sums over $\mathcal{A}$ and $\mathcal{B}$:

$$\begin{aligned} u^{(s^*)} = & \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})] \\ & + \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})] \end{aligned}$$



a | Same-sign mixed bands.

b | Opposite-sign mixed bands.

Figure 3.5 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eqns. (3.37), (3.35).

Consider the sum over $\mathcal{B}$. Looking at the bottom-left panel of Fig. 3.3, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \quad (3.35)$$

A graphical rendition of this idea is given in Fig. 3.5a. Then we can rewrite $u^{(s^*)}$ as a single sum over $\mathcal{A}$,

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) \times [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})] \quad (3.36)$$

From Eq. (3.30), $\forall \mathbf{k} \in \mathcal{A}$ the form factors $\cos k_x \pm \cos k_y$ are positive by construction. By hypothesis $\epsilon^{(s^*)} \geq 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\geq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\geq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \geq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \geq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

This condition, inserted in Eq. (3.36), leads to an absurd:

$$u^{(s^*)} \geq 2t/V \quad u^{(d)} > 0 \quad \implies \quad u^{(s^*)} \leq 0$$

2. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to show that

$$u^{(s^*)} \geq 2t/V > 0 \quad u^{(d)} < 0 \quad \implies \quad u^{(s^*)} \leq 0$$

3. The two points above imply that it is possible to have $u^{(s^*)} \geq 2t/V$ if and only if $u^{(d)} = 0$. But from Sec. 3.4.1, we conclude that the upper boundary $u^{(s^*)} < 2t/V$ always holds.

4. Let for instance

$$u^{(s^*)} \leq 0 \quad u^{(d)} > 0 \quad \implies \quad \epsilon^{(s^*)} < 0 \quad \epsilon^{(d)} > 0$$

In Fig. 3.5b an example band is shown for $\epsilon^{(s^*)} = -1$, $\epsilon^{(d)} = 1/2$.

Proceed as for the first point. Similarly to Eq. (3.35), Any point in $\mathcal{A}$ can be obtained by rotating some point in $\mathcal{B}$,

$$\forall \mathbf{k}' \in \mathcal{A} \quad \exists \quad \mathbf{k} \in \mathcal{B} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \quad (3.37)$$

An equation similar to Eq. (3.36) holds, the sum being performed over $\mathcal{B}$

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) \times [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})] \quad (3.38)$$

From Eq. (3.31), $\forall \mathbf{k} \in \mathcal{B}$ the form factor $\cos k_x + \cos k_y$ is positive by construction; the form factor $\cos k_x - \cos k_y$ is negative by construction. By hypothesis $\epsilon^{(s^*)} < 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\leq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\leq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\leq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\leq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \leq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

This condition, inserted in Eq. (3.36), leads to an absurd:

$$u^{(s^*)} \leq 0 \quad u^{(d)} > 0 \quad \implies \quad u^{(s^*)} > 0$$

5. can be carried out to show that

$$u^{(s^*)} \leq 0 \quad u^{(d)} < 0 \quad \implies \quad u^{(s^*)} > 0$$

6. The two points above imply that it is possible to have $u^{(s^*)} \leq 0$ if and only if $u^{(d)} = 0$. But from Sec. 3.4.1, we conclude that the lower boundary $u^{(s^*)} > 0$ always holds.

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (3.34),

$$0 < u^{(s^*)} < 2t/V$$

holds. In the s^* -wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, “bands shrinking”. What is missing is the behaviour of the d -wave channel when said effect takes place. Continuing the enumeration above,

7. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. Then $\epsilon^{(d)} > 0$. The self-consistency equation for $u^{(d)}$ reads

$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{NBZ}} (\cos k_x - \cos k_y) [f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu})] \end{aligned}$$

where we recognised, as is shown in Fig. 3.4b, that $\text{BZ} \setminus \text{NBZ} = \mathcal{R}\text{NBZ}$. It holds:

$$\forall \mathbf{k} \in \text{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \end{aligned}$$

thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \implies \quad u^{(d)} < 0$$

which is absurd.

8. An identical argument, with inverted inequalities, can be carried out to show that

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \implies \quad u^{(d)} > 0$$

9. The two points above imply that it is possible to satisfy the constraint $0 < u^{(s^*)} < 2t/V$ if and only if $u^{(d)} = 0$.

All the deductions above indicate that $u^{(d)} = 0$ everywhere. Nematic effects are always absent,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \quad (3.39)$$

This is physically expected but non trivial. The normal phase does not break C_4 and we can ignore $u^{(d)}$ at all. In next section we identify the expected self-consistent value for $u^{(s^*)}$.

3.4.3 Expected dependence on δ at low temperature

We understood that $u^{(d)} = 0$. Moreover, the unique HFP $u^{(s^*)}$ is independent of U , but in general it will depend in some way on the doping level. We work at low-temperature, $\beta \rightarrow \infty$. Consider the self-consistency equation of Tab. 3.2 as a function of temperature and chemical potential,

$$u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

in the zero-temperature limit, as long as the band is not flat, it holds

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq \tilde{\mu}} (\cos k_x + \cos k_y)$$

Let $\text{FS}(\delta)$ be the Fermi surface for a given doping level δ . We denote with $\text{Int}(\text{FS})$ be its internal part, the Fermi sea:

$$\begin{aligned} \text{FS} &= \{\mathbf{k} \in \text{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} = \tilde{\mu}\} \\ \text{Int}(\text{FS}) &= \{\mathbf{k} \in \text{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} \leq \tilde{\mu}\} \end{aligned}$$

At half filling, the Fermi surface coincides with the boundary of the MBZ:

$$\delta = 0 \quad \implies \quad \text{FS} = \partial \text{MBZ}$$

This is implied by the fact that renormalised bands are antisymmetric by π shifts,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = -\tilde{\epsilon}_{\mathbf{k}+\pi\sigma}$$

The portion of an s^* -wave band outside the MBZ is mirrored with respect to the portion inside the MBZ. This implies that half the states lie in the MBZ, half out.

At $\delta > 0$ the FS expands in the region $\text{BZ} \setminus \text{MBZ}$ where the form factor $\cos k_x + \cos k_y$ becomes negative. This gives the self-consistency equation

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}[\beta, \tilde{\mu}] \simeq \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq 0} |\cos k_x + \cos k_y| - \frac{1}{L_x L_y} \sum_{0 < \epsilon_{\mathbf{k}} \leq \tilde{\mu}} |\cos k_x + \cos k_y|$$

which, being the sum of a positive and a negative term, is maximised whenever $\tilde{\mu} = 0$, at half filling. This holds as long as the band is not flat. Indeed, if the effective hopping is not zero, to rise the doping level means to

$$\delta_1 > \delta_2 \implies \text{Int}(\text{FS}_1) \subset \text{Int}(\text{FS}_2)$$

with obvious meaning of the notation. Using this property we conclude

$$\tilde{t} > 0 \quad \text{and} \quad \delta_1 > \delta_2 \implies \lim_{\beta \rightarrow \infty} u^{(s^*)}[\beta, \tilde{\mu}] \Big|_{\delta_1} < \lim_{\beta \rightarrow \infty} u^{(s^*)}[\beta, \tilde{\mu}] \Big|_{\delta_2}$$

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is not strong enough to flatten the band, a rise in the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment ($V = 0, \delta$) is included, since on said region of parametric space the model is just the common free electronic model.

Following this line of reasoning, we can extract the two low-temperature limits of $u^{(s^*)}$ at half-filling and unitary filling.

1. At half-filling, the self-consistency equation reduces to

$$u^{(s^*)}[\beta, 0] \Big|_{\beta \gg 1} \simeq \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y)$$

In thermodynamic limit, performing integration on the MBZ shown in the top-left panel of Fig. 3.3:

$$\begin{aligned} \lim_{\beta \rightarrow \infty} u^{(s^*)}[\beta, 0] &\simeq \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} dk_y \int_{-\pi+|k_y|}^{\pi-|k_y|} dk_x \cos k_x + \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} dk_x \int_{-\pi+|k_x|}^{\pi-|k_x|} dk_y \cos k_y \\ &= \frac{1}{2\pi^2} \int_{-\pi}^0 dk_y \int_{-\pi-k_y}^{\pi+k_y} dk_x \cos k_x + \frac{1}{2\pi^2} \int_0^{\pi} dk_y \int_{-\pi+k_y}^{\pi-k_y} dk_x \cos k_x \\ &= \frac{1}{\pi^2} \int_{-\pi}^0 dk_y \sin(\pi + k_y) + \frac{1}{\pi^2} \int_0^{\pi} dk_y \sin(\pi - k_y) \\ &= \left(\frac{2}{\pi}\right)^2 \approx 0.405 \end{aligned} \tag{3.40}$$

2. At unitary filling we have

$$\lim_{\beta \rightarrow \infty} u^{(s^*)}[\beta, 2\tilde{t}^{(s^*)}] = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) = 0$$

which vanishes due to the structure factor periodicity by π .

To summarise: around $V = 0$, $u^{(s^*)}$ is expected to descend monotonically from $4/\pi^2$ to 0 at increasing doping.

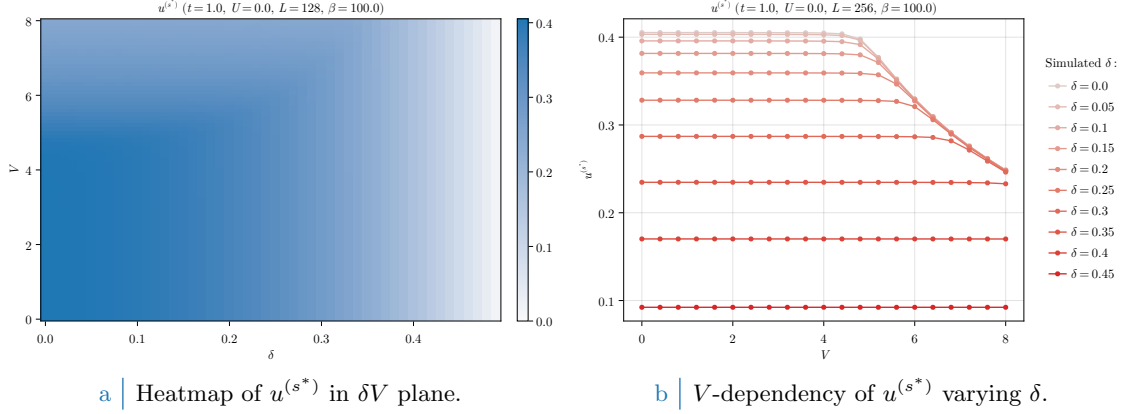


Figure 3.6 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V -independence.

3.5 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. Let us start by sketching the algorithm used to determine self-consistently the mean-field structure of the normal phase in parameter space. For the normal phase, the HFPs “vector” defined in Sec. ?? has just one component:

$$\mathbf{v} = [u^{(s^*)}]$$

As before, we treat spin as a pure degeneracy. The only HFP is characterised by the first self-consistency equation of Tab. 3.2. The resulting model has two RMPs:

$$\begin{aligned} \tilde{t}_{ij} &\equiv \left(t - \frac{u^{(s^*)}V}{2} \right) \varphi_{ij}^{(s^*)} \\ \tilde{\mu} &\equiv \mu + n(2zV - U) \end{aligned}$$

the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (3.27). The normal phase is completely independent of U . Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, with the setup:

$$U = 0 \quad \Delta n = 10^{-2} \quad \Delta \mathbf{v} = [u^{(s^*)} : 5 \times 10^{-4}]$$

with $\Delta \mathbf{v}$ the tolerance vector defined as in Sec. ??.

In Fig. 3.6a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and “low” temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0, 0.1, 0.2, \dots, 4.0 \quad \delta = 0, 0.01, 0.02, \dots, 0.49$$

The expectations of Sec. 3.4.3 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 3.6a. This effect is more visible in Fig. 3.6b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 3.4.3, the half-filled convergence value of $u^{(s^*)}$ in the V -independent region is approximately given by Eq. (3.40).

This effect can only be interpreted in one way, as we discuss in next section: $u^{(s^*)}$ maintains a constant value at fixed doping, as is described in Sec. 3.4.3, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localisation effect.

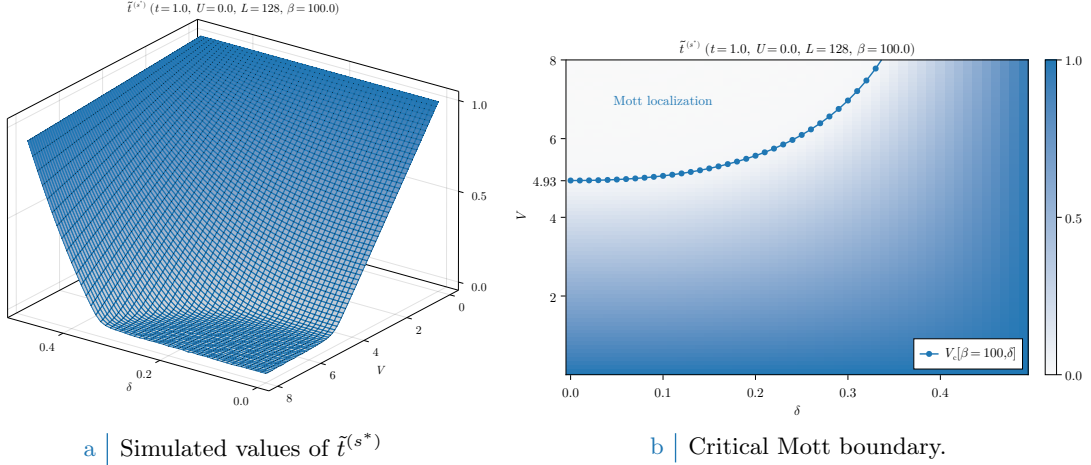


Figure 3.7 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c , computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (3.41). For graphic reasons the points at $V_c > 8$ have been omitted.

3.5.1 Mott localisation effect

The most interesting result we obtained for the normal phase is the possibility of localising electrons by tuning the V/δ ratio. In Fig. 3.6b, at low V the convergence value of $u^{(s^*)}$ is approximately V -independent at sufficiently low temperatures. Let $V_c[\beta, \delta]$ be the critical value such that

$$0 \leq V < V_c[\beta, \delta] \quad : \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

Then within the entire segment $[0, V_c]$ the convergence value is (approximately) the same obtained for $V = 0$. We can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta, \delta] \equiv \frac{2t}{u^{(s^*)}[\beta, \tilde{\mu}(\delta)]|_{V=0}}$$

When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (3.34) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (3.34) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 3.7a the converged values of $\tilde{t}^{(s^*)}$, extracted from the same data of Fig. 3.6a, are plotted in parametric space.

To extract the value of $V_c[\beta, \delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature. Instead, for the high- β half-filled case we can use Eq. (3.40) and get:

$$V_c[\beta \rightarrow \infty, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \quad (3.41)$$

At half-filling, the effective bands amplitude reduction is so strong that small values of V (compared to t) are sufficient to nullify kinetics. In Fig. 3.7b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ computed with the simple procedure hereby described. Our prediction of the critical value for V approximates reasonably the localisation region.

Increasing temperature Mott localisation is suppressed. Also, our precedent argument were valid only at low temperature. In Fig. 3.8a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

$$1.0 \leq \beta \leq 100.0 \quad \delta = 0.0, 0.2, 0.4$$

At high temperature the V -independence region that was visible in Fig. 3.6b completely disappears, making $u^{(s^*)}$ a monotonically descendent function. In Fig. 3.8a the same dataset is plotted with

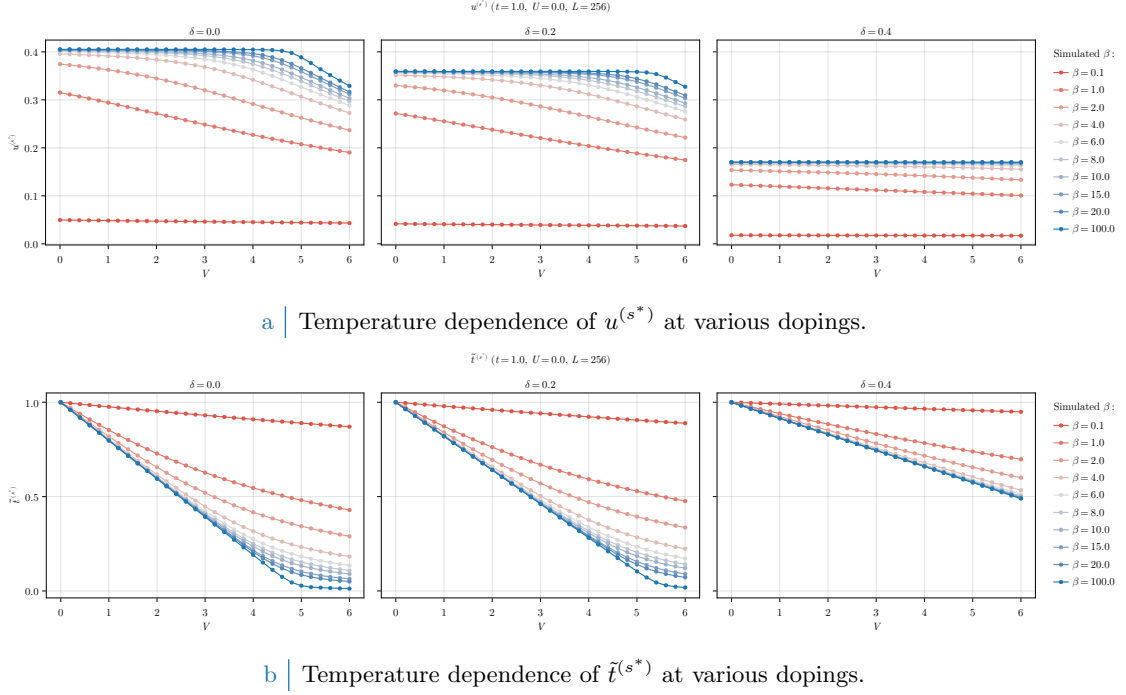


Figure 3.8 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), “medium-doped” ($\delta = 0.2$, central plots) and “heavily-doped” ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localisation, reducing the bands shrinking effect.

focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localisation effects, at least at small values of V . However, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

3.5.2 Solution free energy and entropy densities

In order to compare phases, we need an estimation of free energy of each. The plots of Fig. 3.9 report our findings on the normal phase free energy, computed based on Eq. (3.27). The free energy is intended in unities of t . For a non-flat band, doping increases energy because the band fills up and electrons populate higher energy levels.

Looking the data of Fig. 3.9a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present, the HFP correction of Eq. (3.27) compensates the free energy increase due to bands shrinking,

$$\begin{aligned} -V|u^{(s^*)}|^2 &= -\frac{Vu^{(s^*)*}}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

having we used in the second passage

$$\tilde{\epsilon}_{\mathbf{k}} = \tilde{\epsilon}_{\mathbf{k}} + u^{(s^*)*} V (\cos k_x + \cos k_y)$$

Recalling the first point of the derivation of Sec. 3.3, to add the HFP correction to free energy is the same as removing the *tilda* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.

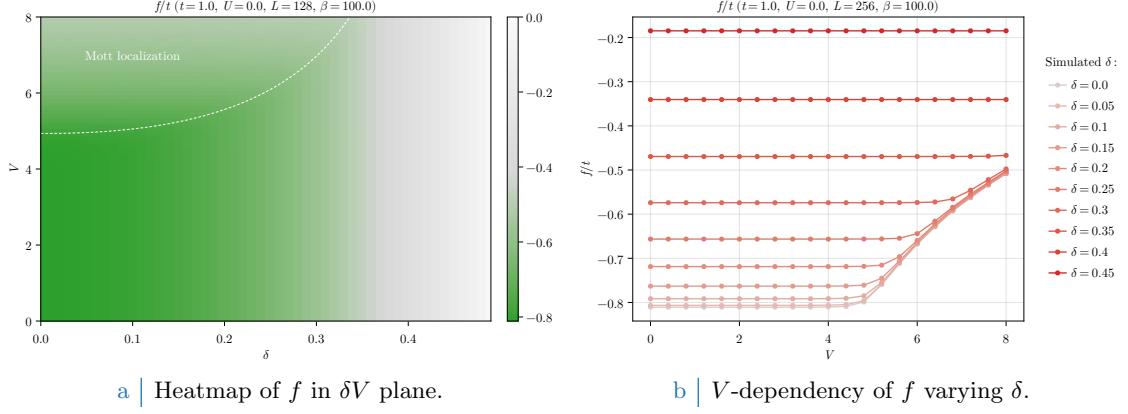


Figure 3.9 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 3.7b, indicating the Mott transition.

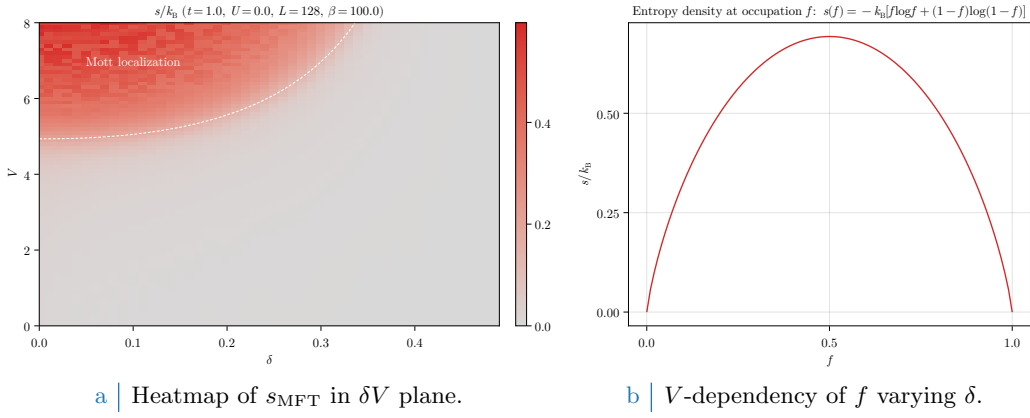


Figure 3.10 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f . On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 3.7b, indicating the Mott transition.

Mott localisation is energetically expensive. This is visible in both panels of Fig. 3.9. The V -dependent energy increase is an entropic effect: indeed, for a flat band:

$$f(0; \beta, \tilde{\mu}(\delta)) = \frac{1}{2}$$

at any temperature. The expression for entropy density of a free Fermi gas [29] is given by

$$s(f) = -k_B [f \log f + (1 - f) \log(1 - f)]$$

represented in Fig. 3.10b. The maximum is located at $f = 1/2$, then a flat band realises the maximally entropic condition. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β , finite-temperature effects become more and more relevant and the statements above need to be corrected.

3.5.3 Convergence problems and mixing fine-tuning

The normal phase is simple enough to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. ?? . Bands flattening is a source of algorithmic instability: numerics can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (3.34). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract the actual convergence values, it is necessary to optimise the mixing parameter $g \in [0, 1]$.

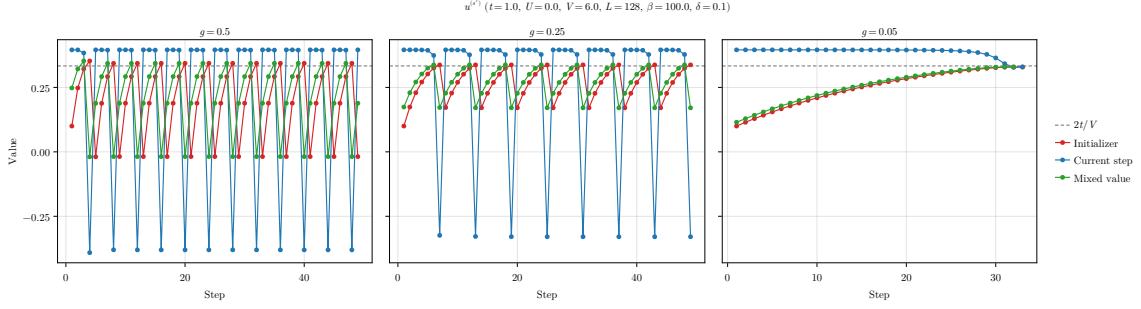


Figure 3.11 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a “Mott-deep” point.

We defined g in Sec. ??, and operatively in Eq. (??). This parameter controls deterministically how, from an initialiser (the HFPs vector we plug in the algorithm) and an output (the HFPs vector we get back from the algorithm) the new initialiser for the following step is generated. We show in Fig. 3.11 the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \quad \delta = 0.1 \quad \beta = 100.0$$

This point is inside the Mott region, as can be checked in Fig. 3.6a. The HFP $u^{(s^*)}$ needs to converge at $2t/V = t/3$ in order to give complete localisation. The initialiser

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

was arbitrarily chosen. In each panel of Fig. 3.11, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue the result of the self-consistent equation, the green one the mixed value of the two based on different g s. The gray dashed line is the theoretical convergence target.

In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. At those steps, the green point is out of bounds with respect to Eq. (3.34). When this happens, the following red point gives a band with inverted sign, a feature that “breaks” the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move “slowly”. As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence. In the track, each blue dot is above the target and would produce a band with inverted sign. Mixing blue and red dots with a 95% unbalance on the red ones, we ensure that the green track approaches the target from below. Note, however, that lowering g cannot be reiterated too many times. Indeed, for $g \rightarrow 0$, any point of our map becomes a fixed point.

This kind of problems can turn out to be pretty common for iHF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

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